CHAPTER 26
WIND LOADS: GENERAL REQUIREMENTS

26.1 PROCEDURES

26.1.1 Scope. Buildings and other structures, including the main wind force resisting system (MWFRS) and all components and cladding (C&C) thereof, shall be designed and constructed to resist the wind loads determined in accordance with Chapters 26 through 31. The provisions of this chapter define basic wind parameters for use with other provisions contained in this standard.

26.1.2 Permitted Procedures. The design wind loads for buildings and other structures, including the MWFRS and C&C elements thereof, shall be determined using one of the procedures as specified in this section. An outline of the overall process for the determination of the wind loads, including section references, is provided in Fig. 26.1-1.

26.1.2.1 Main Wind Force Resisting System. Wind loads for the MWFRS shall be determined using one of the following procedures:

1. Directional Procedure for buildings of all heights as specified in Chapter 27 for buildings meeting the requirements specified therein;
2. Envelope Procedure for low-rise buildings as specified in Chapter 28 for buildings meeting the requirements specified therein;
3. Directional Procedure for Building Appurtenances (rooftop structures and rooftop equipment) and Other Structures (such as solid freestanding walls and solid freestanding signs, chimneys, tanks, open signs, single-plane open frames, and trussed towers) as specified in Chapter 29; or
4. Wind Tunnel Procedure for all buildings and all other structures as specified in Chapter 31.

26.1.2.2 Components and Cladding. Wind loads on C&C on all buildings and other structures shall be designed using one of the following procedures:

1. Analytical Procedures provided in Parts 1 through 6, as appropriate, of Chapter 30; or
2. Wind Tunnel Procedure as specified in Chapter 31.

26.2 DEFINITIONS

The following definitions apply to the provisions of Chapters 26 through 31:

APPROVED: Acceptable to the Authority Having Jurisdiction.

ATTACHED CANOPY: A horizontal (maximum slope of 2%) patio cover attached to the building wall at any height; it is different from an overhang, which is an extension of the roof surface.

BASIC WIND SPEED, *V*: Three-second gust speed at 33 ft (10 m) above the ground in Exposure C (see Section 26.7.3) as determined in accordance with Section 26.5.1.

BUILDING, ENCLOSED: A building that has the total area of openings in each wall, that receives positive external pressure, less than or equal to 4 sq ft (0.37 m²) or 1% of the area of that wall, whichever is smaller. This condition is expressed for each wall by the following equation:

$$A_o < 0.01A_g,\ \text{or}\ 4\ \text{sq ft}\ (0.37\ \text{m}^2),\ \text{whichever is smaller},$$

where A_o and A_g are as defined for Open Buildings.

BUILDING, LOW-RISE: Enclosed or partially enclosed building that complies with the following conditions:

1. Mean roof height h less than or equal to 60 ft (18 m).
2. Mean roof height h does not exceed least horizontal dimension.

BUILDING, OPEN: A building that has each wall at least 80% open. This condition is expressed for each wall by the equation $A_o \geq 0.8A_g$, where

A_o = total area of openings in a wall that receives positive external pressure, in ft^2 (m^2); and
A_g = the gross area of that wall in which A_o is identified, in ft^2 (m^2).

BUILDING, PARTIALLY ENCLOSED: A building that complies with both of the following conditions:

1. The total area of openings in a wall that receives positive external pressure exceeds the sum of the areas of openings in the balance of the building envelope (walls and roof) by more than 10%.
2. The total area of openings in a wall that receives positive external pressure exceeds 4 ft^2 (0.37 m^2) or 1% of the area of that wall, whichever is smaller, and the percentage of openings in the balance of the building envelope does not exceed 20%.

These conditions are expressed by the following equations:

$$A_o > 1.10A_{oi}$$

$$A_o > 4\ \text{ft}^2 (0.37\ \text{m}^2)\ \text{or}$$
$$> 0.01A_g,\ \text{whichever is smaller, and}\ A_{oi}/A_{gi} \leq 0.20$$

where A_o and A_g are as defined for Open Building;

A_{oi} = sum of the areas of openings in the building envelope (walls and roof) not including A_o, in ft^2 (m^2); and
A_{gi} = sum of the gross surface areas of the building envelope (walls and roof) not including A_g, in ft^2 (m^2).

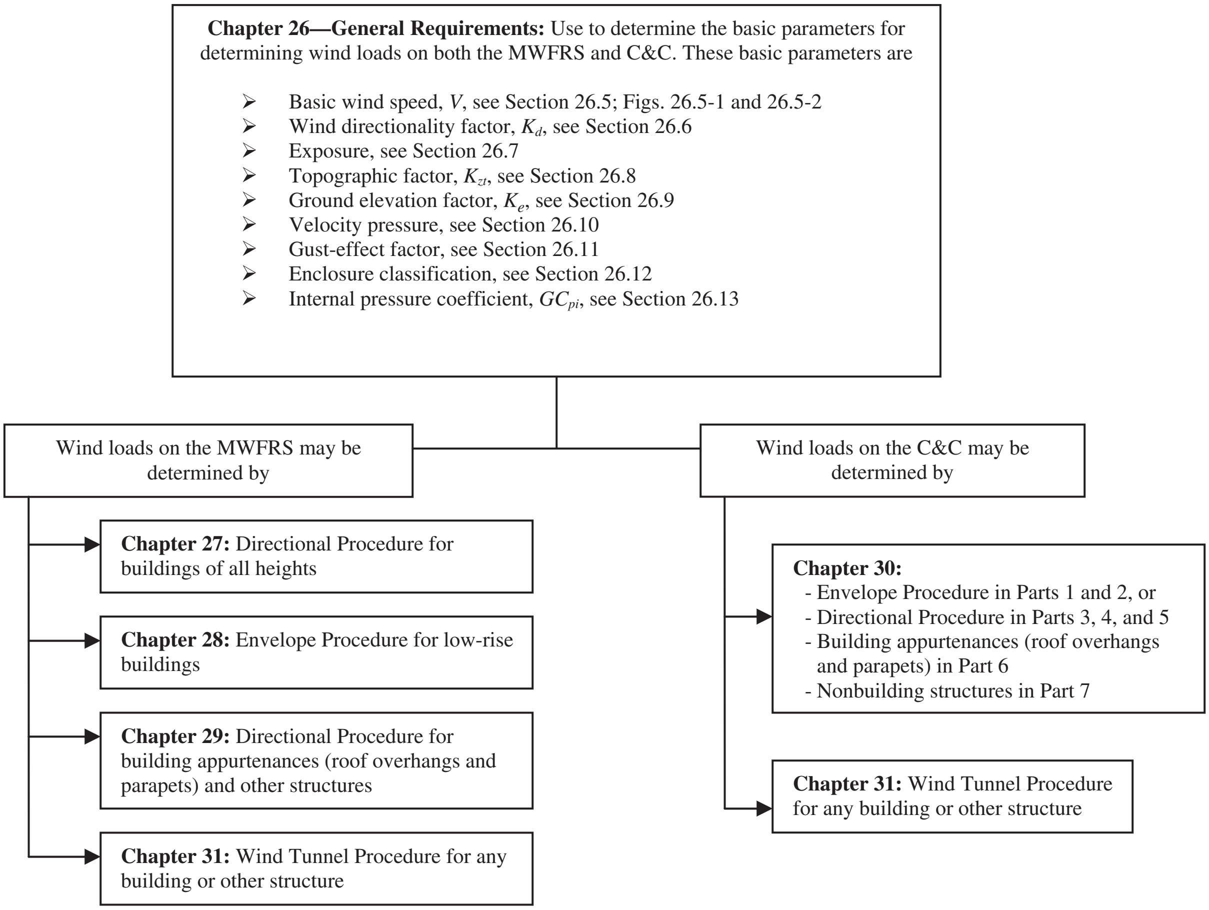


FIGURE 26.1-1 Outline of Process for Determining Wind Loads

Additional outlines and User Notes are provided at the beginning of each chapter for more detailed step-by-step procedures for determining the wind loads.

BUILDING, PARTIALLY OPEN: A building that does not comply with the requirements for open, partially enclosed, or enclosed buildings.

BUILDING, SIMPLE DIAPHRAGM: A building in which both windward and leeward wind loads are transmitted by roof and vertically spanning wall assemblies, through continuous floor and roof diaphragms, to the MWFRS.

BUILDING, TORSIONALLY REGULAR UNDER WIND LOAD: A building with the MWFRS about each principal axis proportioned so that the maximum displacement at each story under Case 2, the torsional wind load case, of Fig. 27.3-8 does not exceed the maximum displacement at the same location under Case 1 of Fig. 27.3-8, the basic wind load case.

BUILDING ENVELOPE: Cladding, roofing, exterior walls, glazing, door assemblies, window assemblies, skylight assemblies, and other components enclosing the building.

BUILDING OR OTHER STRUCTURE, FLEXIBLE: Slender buildings and other structures that have a fundamental natural frequency less than 1 Hz.

BUILDING OR OTHER STRUCTURE, REGULAR-SHAPED: A building or other structure that has no unusual geometrical irregularity in spatial form.

BUILDING OR OTHER STRUCTURE, RIGID: A building or other structure whose fundamental frequency is greater than or equal to 1 Hz.

COMPONENTS AND CLADDING (C&C): Elements of the building envelope or elements of building appurtenances and rooftop structures and equipment that do not qualify as part of the MWFRS.

DESIGN FORCE, F: Equivalent static force to be used in the determination of wind loads for other structures.

DESIGN PRESSURE, p: Equivalent static pressure to be used in the determination of wind loads for buildings.

DIAPHRAGM: Roof, floor, or other membrane or bracing system acting to transfer lateral forces to the vertical MWFRS. For analysis under wind loads, diaphragms constructed of untopped steel decks, concrete-filled steel decks, and concrete slabs, each having a span-to-depth ratio of 2 or less, shall be permitted to be idealized as *rigid*. Diaphragms constructed of wood structural panels are permitted to be idealized as *flexible*.

DIRECTIONAL PROCEDURE: A procedure for determining wind loads on buildings and other structures for specific wind directions, in which the external pressure coefficients used are based on past wind tunnel testing of prototypical building models for the corresponding direction of wind.

EAVE HEIGHT, h_e: The distance from the ground surface adjacent to the building to the roof eave line at a particular wall. If the height of the eave varies along the wall, the average height shall be used.

EFFECTIVE WIND AREA, A: The area used to determine the external pressure coefficient, (GC_p) and (GC_{rn}). For C&C elements, the effective wind area in Figs. 30.3-1 through 30.3-7, 30.4-1, 30.5-1, and 30.7-1 through 30.7-3 is the span length multiplied by an effective width that need not be less than one-third the span length. For rooftop solar arrays, the effective wind area in Fig. 29.4-7 is equal to the tributary area for the structural element being considered, except that the width of the effective wind area need not be less than one-third its length. For cladding fasteners, the effective wind area shall not be greater than the area that is tributary to an individual fastener.

ENVELOPE PROCEDURE: A procedure for determining wind load cases on buildings, in which pseudoexternal pressure coefficients are derived from past wind tunnel testing of prototypical building models successively rotated through 360°, such that the pseudopressure cases produce key structural actions (e.g., uplift, horizontal shear, and bending moments) that envelop their maximum values among all possible wind directions.

ESCARPMENT: With respect to topographic effects in Section 26.8, a cliff or steep slope generally separating two levels or gently sloping areas (see Fig. 26.8-1). Also known as a scarp.

FREE ROOF: Roof with a configuration generally conforming to those shown in Figs. 27.3-4 through 27.3-6 (monoslope, pitched, or troughed) in an open building with no enclosing walls underneath the roof surface.

GLAZING: Glass or transparent or translucent plastic sheet used in windows, doors, skylights, or curtain walls.

GLAZING, IMPACT-RESISTANT: Glazing that has been shown by testing to withstand the impact of test missiles. See Section 26.12.3.2.

HILL: With respect to topographic effects in Section 26.8, a land surface characterized by strong relief in any horizontal direction (see Fig. 26.8-1).

HURRICANE-PRONE REGIONS: Areas vulnerable to hurricanes; in the United States and its territories, defined as

1. The U.S. Atlantic Ocean and Gulf of Mexico coasts where the basic wind speed for Risk Category II buildings is greater than 115 mi/h (51.4 m/s); and
2. Hawaii, Puerto Rico, Guam, Virgin Islands, and American Samoa.

IMPACT-PROTECTIVE SYSTEM: Construction that has been shown by testing to withstand the impact of test missiles and that is applied, attached, or locked over exterior glazing. See Section 26.12.3.2.

MAIN WIND FORCE RESISTING SYSTEM (MWFRS): An assemblage of structural elements assigned to provide support and stability for the overall building or other structure. The system generally receives wind loading from more than one surface.

MEAN ROOF HEIGHT, *h*: The average of the roof eave height and the height to the highest point on the roof surface, except that, for roof angles less than or equal to 10°, the mean roof height is permitted to be taken as the roof eave height.

OPENINGS: Apertures or holes in the building envelope that allow air to flow through the building envelope and that are designed as "open" during design winds as defined by these provisions.

RECOGNIZED LITERATURE: Published research findings and technical papers that are approved.

RIDGE: With respect to topographic effects in Section 26.8, an elongated crest of a hill characterized by strong relief in two directions (see Fig. 26.8-1).

ROOFTOP SOLAR PANEL: A device to receive solar radiation and convert it into electricity or heat energy. Typically this is a photovoltaic module or solar thermal panel.

SOLAR ARRAY: Any number of rooftop solar panels grouped closely together.

WIND-BORNE DEBRIS REGIONS: Areas within hurricane-prone regions where impact protection is required for glazed openings; see Section 26.12.3.

WIND TUNNEL PROCEDURE: A procedure for determining wind loads on buildings and other structures, in which pressures and/or forces and moments are determined for each wind direction considered, from a model of the building or other structure and its surroundings, in accordance with Chapter 31.

26.3 SYMBOLS

The following symbols apply only to the provisions of Chapters 26 through 31:

A = effective wind area, in ft^2 (m^2)
A_f = area of open buildings and other structures either normal to the wind direction or projected on a plane normal to the wind direction, in ft^2 (m^2)
A_g = gross area of that wall in which A_o is identified, in ft^2 (m^2)
A_{gi} = sum of the gross surface areas of the building envelope (walls and roof) not including A_g, in ft^2 (m^2)
A_n = normalized wind area for rooftop solar panels in Fig. 29.4-7
A_o = total area of openings in a wall that receives positive external pressure, in ft^2 (m^2)
A_{og} = total area of openings in the building envelope in ft^2 (m^2)
A_{oi} = sum of the areas of openings in the building envelope (walls and roof) not including A_o, in ft^2 (m^2)
A_s = gross area of the solid freestanding wall or solid sign, in ft^2 (m^2)
a = width of pressure coefficient zone, in ft (m)
B = horizontal dimension of building measured normal to wind direction, in ft (m)
$\bar{b}$ = mean hourly wind speed factor in Eq. (26.11-16) from Table 26.11-1
$\hat{b}$ = 3-s gust speed factor from Table 26.11-1
c = turbulence intensity factor in Eq. (26.11-7) from Table 26.11-1
C_f = force coefficient to be used in determination of wind loads for other structures
C_N = net pressure coefficient to be used in determination of wind loads for open buildings
C_p = external pressure coefficient to be used in determination of wind loads for buildings
D = diameter of a circular structure or member, in ft (m)
D' = depth of protruding elements such as ribs and spoilers, in ft (m)
d_1 = for rooftop solar arrays, horizontal distance orthogonal to the panel edge to an adjacent panel or the building edge, ignoring any rooftop equipment in Fig. 29.4-7, in ft (m)
d_2 = for rooftop solar arrays, horizontal distance from the edge of one panel to the nearest edge in the next row of panels in Fig. 29.4-7, in ft (m)
F = design wind force for other structures, in lb (N)
G = gust-effect factor
G_f = gust-effect factor for MWFRS of flexible buildings and other structures
(GC_p) = product of external pressure coefficient and gust-effect factor to be used in determination of wind loads for buildings

(GC_{pf}) = product of the equivalent external pressure coefficient and gust-effect factor to be used in determination of wind loads for MWFRS of low-rise buildings

(GC_{pi}) = product of internal pressure coefficient and gust-effect factor to be used in determination of wind loads for buildings

(GC_{pn}) = combined net pressure coefficient for a parapet

(GC_r) = product of external pressure coefficient and gust-effect factor to be used in determination of wind loads for rooftop structures

(GC_{rn}) = net pressure coefficient for rooftop solar panels, in Eqs. (29.4-4) and (29.4-5)

$(GC_{rn})_{\text{nom}}$ = nominal net pressure coefficient for rooftop solar panels determined from Fig. 29.4-7

g_Q = peak factor for background response in Eqs. (26.11-6) and (26.11-10)

g_R = peak factor for resonant response in Eq. (26.11-10)

g_v = peak factor for wind response in Eqs. (26.11-6) and (26.11-10)

H = height of hill, ridge, or escarpment in Fig. 26.8-1, in ft (m)

h = mean roof height of a building or height of other structure, except that eave height shall be used for roof angle θ less than or equal to 10°, in ft (m)

h_1 = height of a solar panel above the roof at the lower edge of the panel, in ft (m)

h_2 = height of a solar panel above the roof at the upper edge of the panel, in ft (m)

h_e = roof eave height at a particular wall, or the average height if the eave varies along the wall

h_p = height to top of parapet in Figs. 27.5-2 and 30.6-1

h_{pt} = mean parapet height above the adjacent roof surface for use with Eq. (29.4-5), in ft (m)

$I_{\bar{z}}$ = intensity of turbulence from Eq. (26.11-7)

K_1, K_2, K_3 = multipliers in Fig. 26.8-1 to obtain K_{zt}

K_d = wind directionality factor in Table 26.6-1

K_e = Ground elevation factor

K_h = velocity pressure exposure coefficient evaluated at height $z = h$

K_z = velocity pressure exposure coefficient evaluated at height z

K_{zt} = topographic factor as defined in Section 26.8

L = horizontal dimension of a building measured parallel to the wind direction, in ft (m)

L_b = normalized building length, for use with Fig. 29.4-7, in ft (m)

L_h = distance upwind of crest of hill, ridge, or escarpment in Fig. 26.8-1 to where the difference in ground elevation is half the height of the hill, ridge, or escarpment, in ft (m)

L_p = panel chord length for use with rooftop solar panels in Fig. 29.4-7, in ft (m)

L_r = horizontal dimension of return corner for a solid freestanding wall or solid sign from Fig. 29.3-1, in ft (m)

$L_{\bar{z}}$ = integral length scale of turbulence, in ft (m)

ℓ = integral length scale factor from Table 26.11-1, ft (m)

N_1 = reduced frequency from Eq. (26.11-14)

n_1 = fundamental natural frequency, in Hz

n_a = approximate lower bound natural frequency (Hz) from Section 26.11.2

p = design pressure to be used in determination of wind loads for buildings, in lb/ft^2 (N/m^2)

P_L = wind pressure acting on leeward face in Fig. 27.3-8, in lb/ft^2 (N/m^2)

p_{net} = net design wind pressure from Eq. (30.4-1), in lb/ft^2 (N/m^2)

p_{net30} = net design wind pressure for Exposure B at $h = 30$ ft (9.1 m) and $I = 1.0$ from Fig. 30.4-1, in lb/ft^2 (N/m^2)

p_p = combined net pressure on a parapet from Eq. (27.3-4), in lb/ft^2 (N/m^2)

p_s = net design wind pressure from Eq. (28.5-1), in lb/ft^2 (N/m^2)

p_{s30} = simplified design wind pressure for Exposure B at $h = 30$ ft (9.1 m) and $I = 1.0$ from Fig. 28.5-1, in lb/ft^2 (N/m^2)

P_W = wind pressure acting on windward face in Fig. 27.3-8, in lb/ft^2 (N/m^2)

Q = background response factor from Eq. (26.11-8)

q = velocity pressure, in lb/ft^2 (N/m^2)

q_h = velocity pressure evaluated at height $z = h$, in lb/ft^2 (N/m^2)

q_i = velocity pressure for internal pressure determination, in lb/ft^2 (N/m^2)

q_p = velocity pressure at top of parapet, in lb/ft^2 (N/m^2)

q_z = velocity pressure evaluated at height z above ground, in lb/ft^2 (N/m^2)

R = resonant response factor from Eq. (26.11-12)

r = rise-to-span ratio for arched roofs

R_B, R_h, R_L = values from Eqs. (26.11-15a) and (26.11-15b)

R_i = reduction factor from Eq. (26.13-1)

R_n = value from Eq. (26.11-13)

s = vertical dimension of the solid freestanding wall or solid sign from Fig. 29.3-1, in ft (m)

V = basic wind speed obtained from Figs. 26.5-1A through 26.5-1D and 26.5-2A through 26.5-2D, in mi/h (m/s). The basic wind speed corresponds to a 3-s gust speed at 33 ft (10 m) above the ground in Exposure Category C

V_i = unpartitioned internal volume, in ft^3 (m^3)

$\bar{V}_{\bar{z}}$ = mean hourly wind speed at height $\bar{z}$, in ft/s (m/s)

W = width of building in Figs. 30.3-3, 30.3-5A, and 30.3-5B and width of span in Figs. 30.3-4 and 30.3-6, in ft (m)

W_L = width of a building on its longest side in Fig. 29.4-7, in ft (m)

W_S = width of a building on its shortest side in Fig. 29.4-7, in ft (m)

x = distance upwind or downwind of crest in Fig. 26.8-1, in ft (m)

z = height above ground level, in ft (m)

$\bar{z}$ = equivalent height of structure, in ft (m)

z_g = nominal height of the atmospheric boundary layer used in this standard (values appear in Table 26.11-1)

$z_{\min}$ = exposure constant from Table 26.11-1

α = 3-s gust-speed power law exponent from Table 26.11-1

$\hat{\alpha}$ = reciprocal of α from Table 26.11-1

$\bar{\alpha}$ = mean hourly wind-speed power law exponent in Eq. (26.11-16) from Table 26.11-1

β = damping ratio, percent critical for buildings or other structures

γ_c = panel chord factor for use with rooftop solar panels in Eq. (29.4-5)

γ_E = array edge factor for use with rooftop solar panels in Fig. 29.4-7 and Eqs. (29.4-4) and (29.4-5)

γ_p = parapet height factor for use with rooftop solar panels in Eq. (29.4-5)
ε = ratio of solid area to gross area for solid freestanding wall, solid sign, open sign, face of a trussed tower, or lattice structure
$\bar{\varepsilon}$ = integral length scale power law exponent in Eq. (26.11-9) from Table 26.11-1
η = value used in Eqs. (26.11-15a) and (26.11-15b) (see Section 26.11.4)
θ = angle of plane of roof from horizontal, in degrees
λ = adjustment factor for building height and exposure from Figs. 28.5-1 and 30.4-1
v = height-to-width ratio for solid sign
ω = angle that the solar panel makes with the roof surface in Fig. 29.4-7, in degrees

26.4 GENERAL

26.4.1 Sign Convention. Positive pressure acts toward the surface and negative pressure acts away from the surface.

26.4.2 Critical Load Condition. Values of external and internal pressures shall be combined algebraically to determine the most critical load.

26.4.3 Wind Pressures Acting on Opposite Faces of Each Building Surface. In the calculation of design wind loads for the MWFRS and for C&C for buildings, the algebraic sum of the pressures acting on opposite faces of each building surface shall be taken into account.

26.5 WIND HAZARD MAP

26.5.1 Basic Wind Speed. The basic wind speed, V, used in the determination of design wind loads on buildings and other structures shall be determined from Figs. 26.5-1 and 26.5-2 as follows, except as provided in Sections 26.5.2 and 26.5.3:

For Risk Category I buildings and structures, use Figs. 26.5-1A and 26.5-2A.
For Risk Category II buildings and structures, use Figs. 26.5-1B and 26.5-2B.
For Risk Category III buildings and structures, use Figs. 26.5-1C and 26.5-2C.
For Risk Category IV buildings and structures, use Figs. 26.5-1D and 26.5-2D.

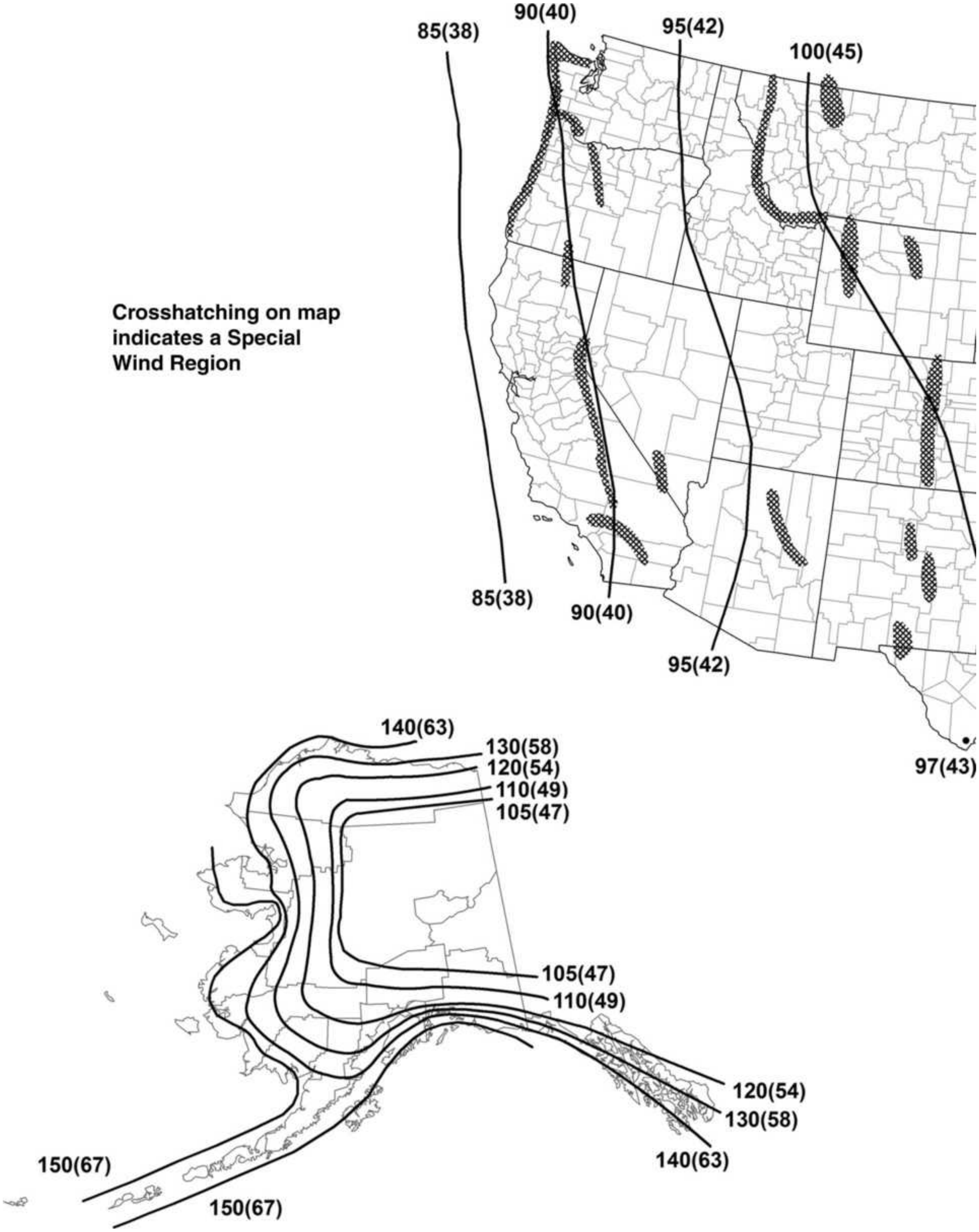
The wind shall be assumed to come from any horizontal direction. The basic wind speed shall be increased where records or experience indicate that the wind speeds are higher than those reflected in Figs. 26.5-1 and 26.5-2.

26.5.2 Special Wind Regions. Mountainous terrain, gorges, and special wind regions shown in Fig. 26.5-1 shall be examined for unusual wind conditions. The Authority Having Jurisdiction shall, if necessary, adjust the values given in Fig. 26.5-1 to account for higher local wind speeds. Such adjustment shall be based on meteorological information and an estimate of the basic wind speed obtained in accordance with the provisions of Section 26.5.3.

26.5.3 Estimation of Basic Wind Speeds from Regional Climatic Data. In areas outside hurricane-prone regions, regional climatic data shall only be used in lieu of the basic wind speeds given in Figs. 26.5-1 and 26.5-2 when (1) approved extreme-value statistical analysis procedures have been used in reducing the data; and (2) the length of record, sampling error, averaging time, anemometer height, data quality, and terrain exposure of the anemometer have been taken into account. Reduction in basic wind speed below that of Figs. 26.5-1 and 26.5-2 shall be permitted.

In hurricane-prone regions, wind speeds derived from simulation techniques shall only be used in lieu of the basic wind speeds given in Figs. 26.5-1 and 26.5-2 when approved simulation and extreme-value statistical analysis procedures are used. The use of regional wind speed data obtained from anemometers is not permitted to define the hurricane wind-speed risk along the Gulf and Atlantic coasts, the Caribbean, or Hawaii.

When the basic wind speed is estimated from regional climatic data or simulation, the estimate shall correspond to the applicable mean recurrence interval, and the estimate shall be adjusted for equivalence to a 3-s gust wind speed at 33 ft (10 m) above ground in Exposure C.



Notes

1. Values are nominal design 3-s gust wind speeds in mi/h (m/s) at 33 ft (10 m) above ground for Exposure Category C.
2. Linear interpolation is permitted between contours. Point values are provided to aid with interpolation.
3. Islands, coastal areas, and land boundaries outside the last contour shall use the last wind speed contour.
4. Mountainous terrain, gorges, ocean promontories, and special wind regions shall be examined for unusual wind conditions.
5. Wind speeds correspond to approximately a 15% probability of exceedance in 50 years (Annual Exceedance Probability = 0.00333, MRI = 300 years).
6. Location-specific basic wind speeds shall be permitted to be determined using www.atcouncil.org/windspeed.

FIGURE 26.5-1A Basic Wind Speeds for Risk Category I Buildings and Other Structures

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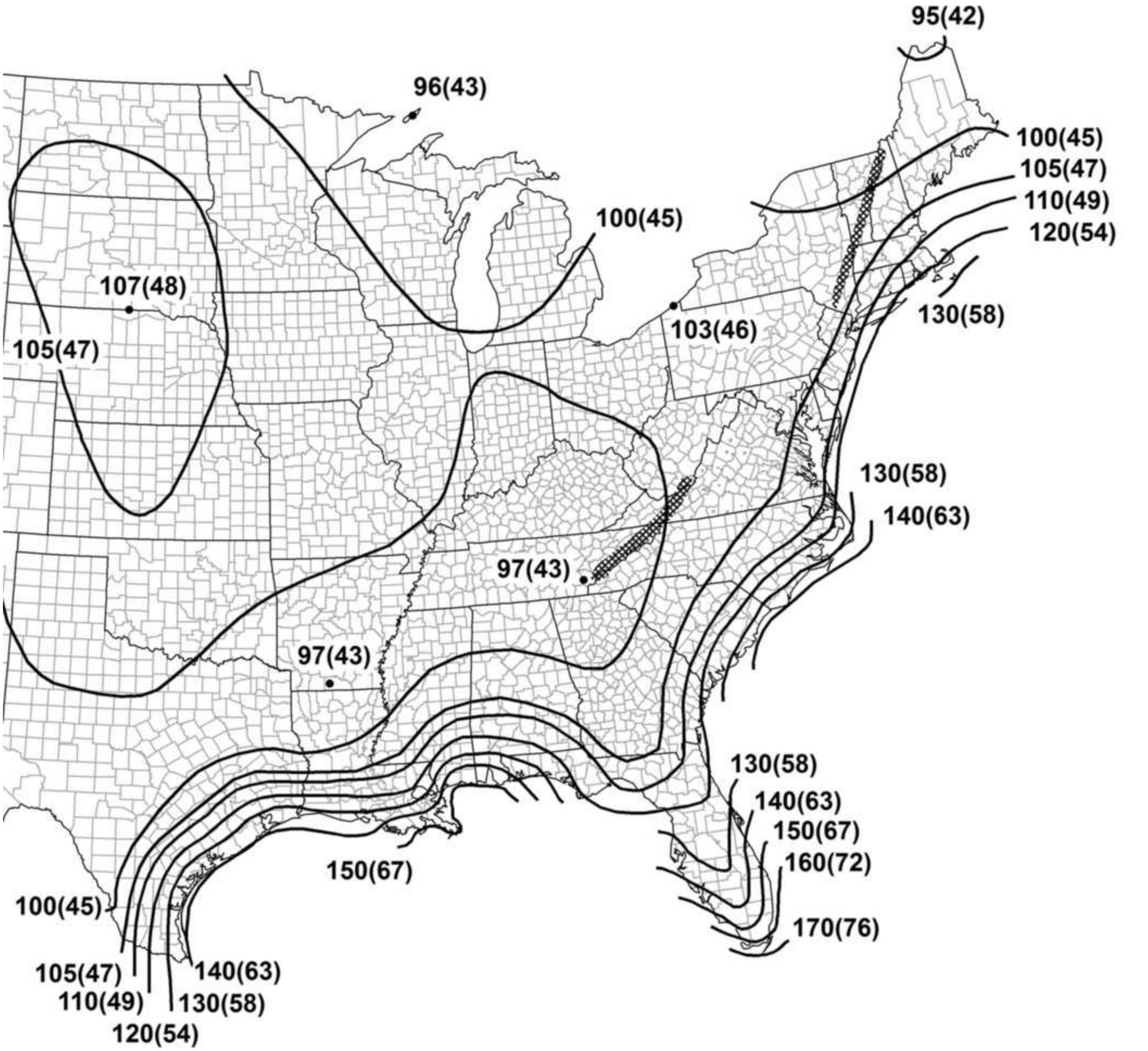


Location	Vmph	(m/s)
Guam	180	(80)
Virgin Islands	150	(67)
American Samoa	150	(67)
Hawaii	(See Figure 26.5-2A)	

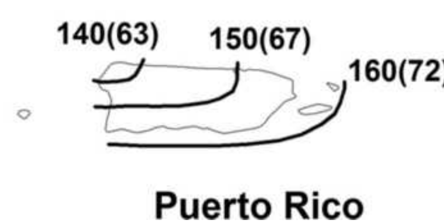
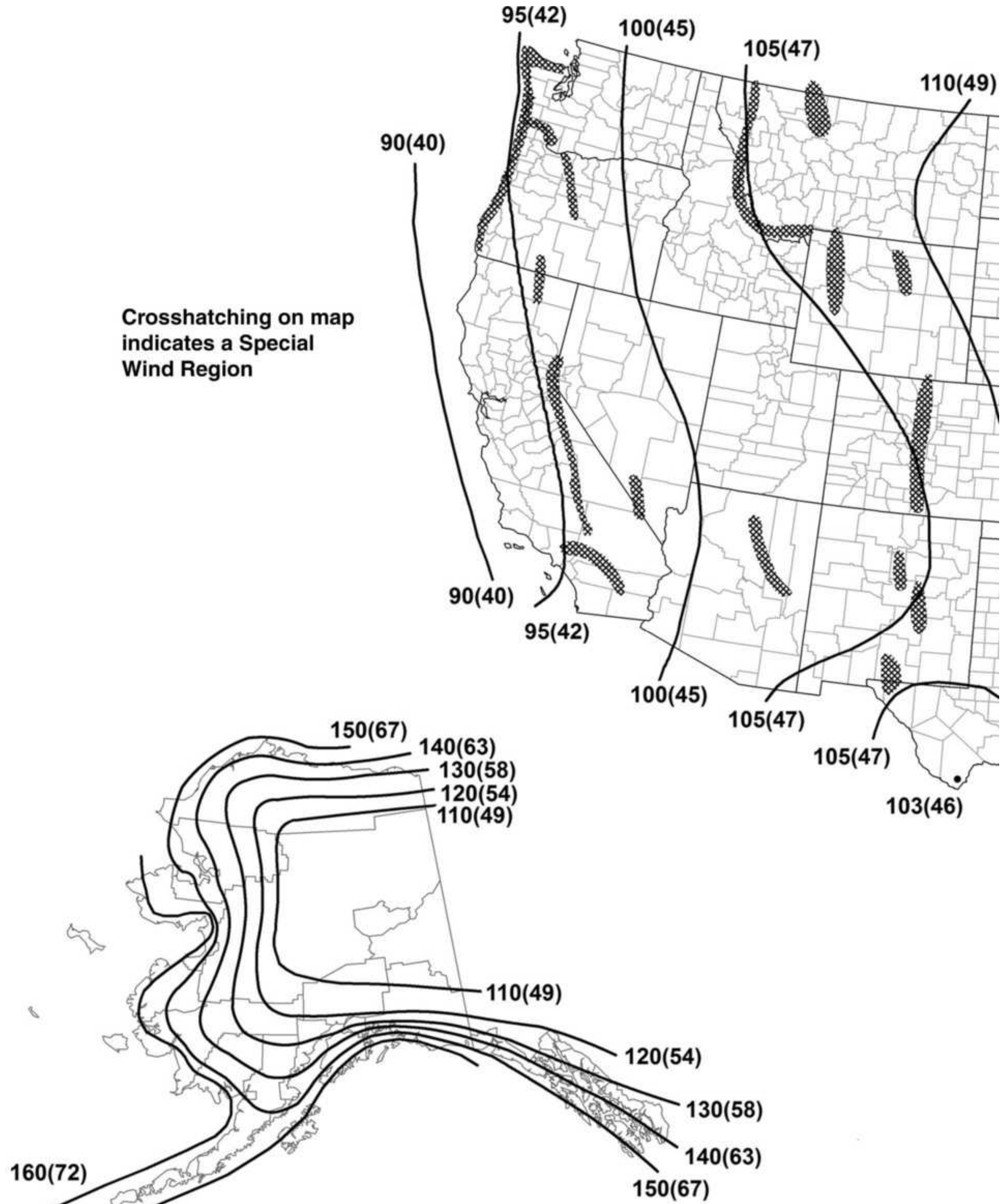


FIGURE 26.5-1A (*Continued*). Basic Wind Speeds for Risk Category I Buildings and Other Structures



Notes

1. Values are nominal design 3-s gust wind speeds in mi/h (m/s) at 33 ft (10 m) above ground for Exposure Category C.
2. Linear interpolation is permitted between contours. Point values are provided to aid with interpolation.
3. Islands, coastal areas, and land boundaries outside the last contour shall use the last wind speed contour.
4. Mountainous terrain, gorges, ocean promontories, and special wind regions shall be examined for unusual wind conditions.
5. Wind speeds correspond to approximately a 7% probability of exceedance in 50 years (Annual Exceedance Probability = 0.00143, MRI = 700 years).
6. Location-specific basic wind speeds shall be permitted to be determined using www.atcouncil.org/windspeed.

FIGURE 26.5-1B Basic Wind Speeds for Risk Category II Buildings and Other Structures

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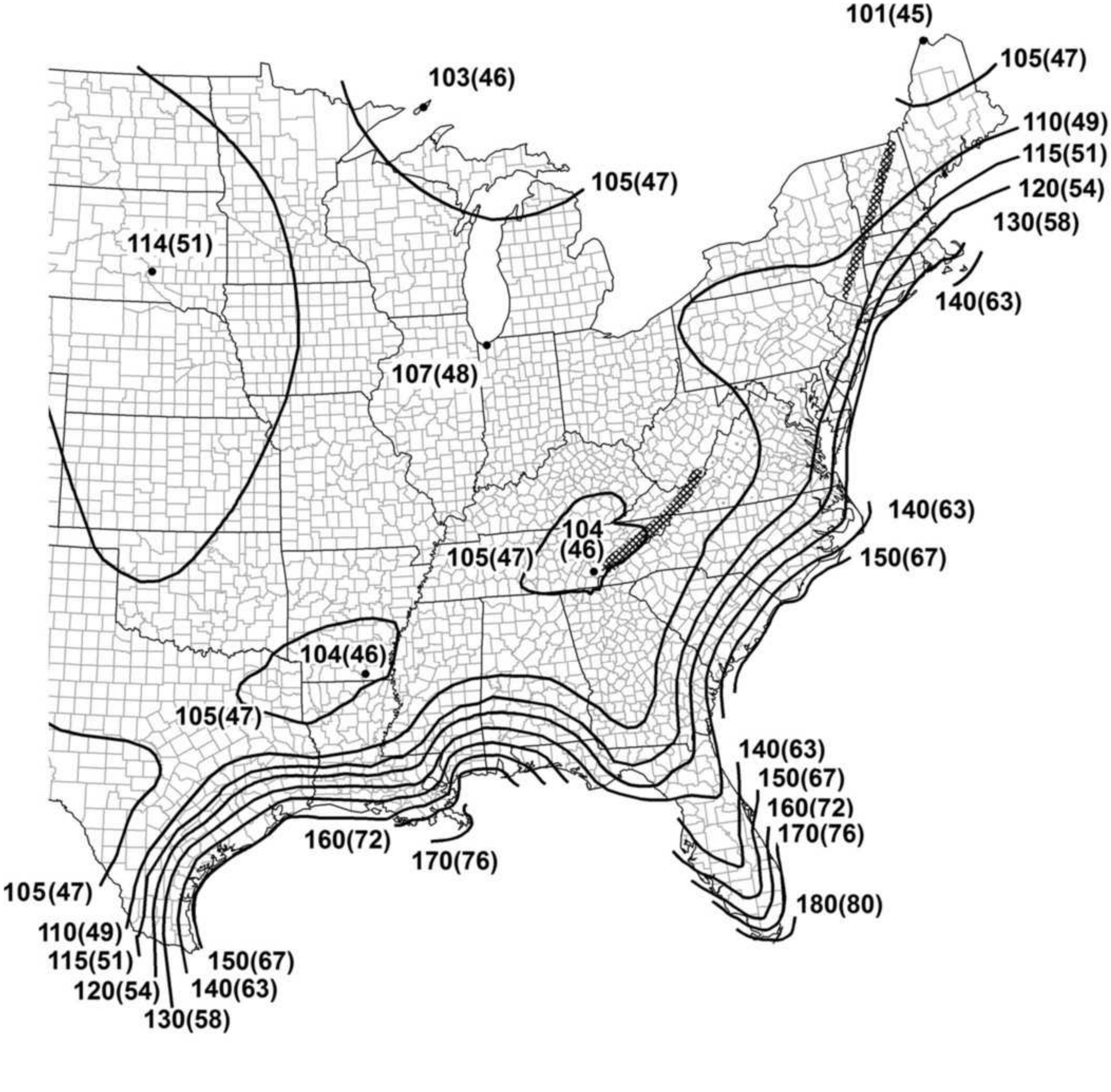


Location	V (mph)	V (m/s)
Guam	195	(87)
Virgin Islands	165	(74)
American Samoa	160	(72)
Hawaii	See Figure 26.5-2B	

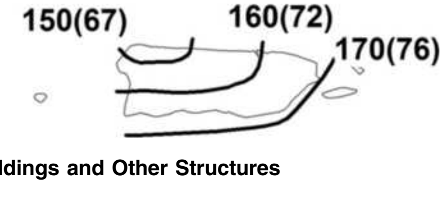
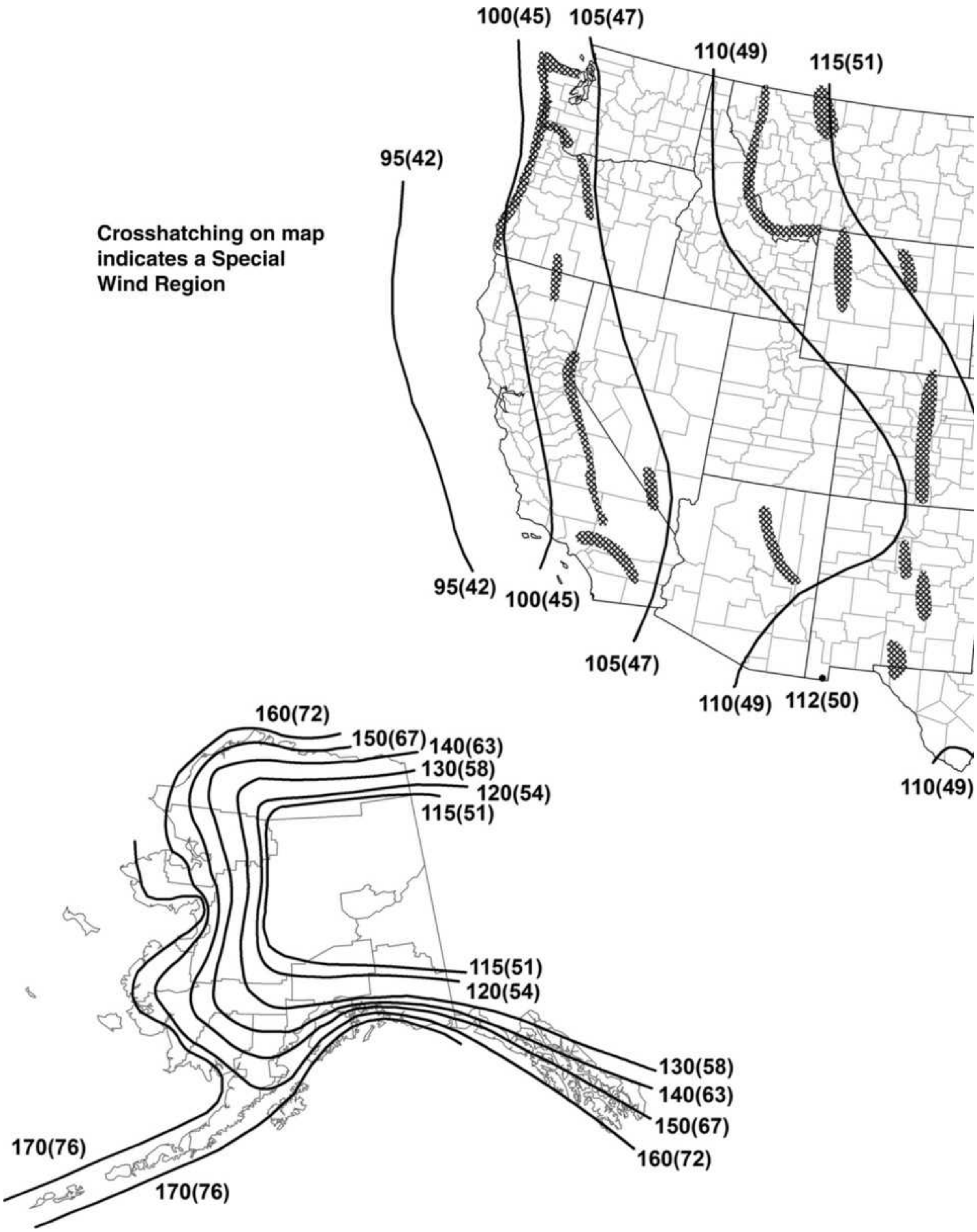


FIGURE 26.5-1B (*Continued*). Basic Wind Speeds for Risk Category II Buildings and Other Structures



Notes

1. Values are nominal design 3-s gust wind speeds in mi/h (m/s) at 33 ft (10 m) above ground for Exposure Category C.
2. Linear interpolation is permitted between contours. Point values are provided to aid with interpolation.
3. Islands, coastal areas, and land boundaries outside the last contour shall use the last wind speed contour.
4. Mountainous terrain, gorges, ocean promontories, and special wind regions shall be examined for unusual wind conditions.
5. Wind speeds correspond to approximately a 3% probability of exceedance in 50 years (Annual Exceedance Probability = 0.000588, MRI = 1,700 years).
6. Location-specific basic wind speeds shall be permitted to be determined using www.atcouncil.org/windspeed.

FIGURE 26.5-1C Basic Wind Speeds for Risk Category III Buildings and Other Structures

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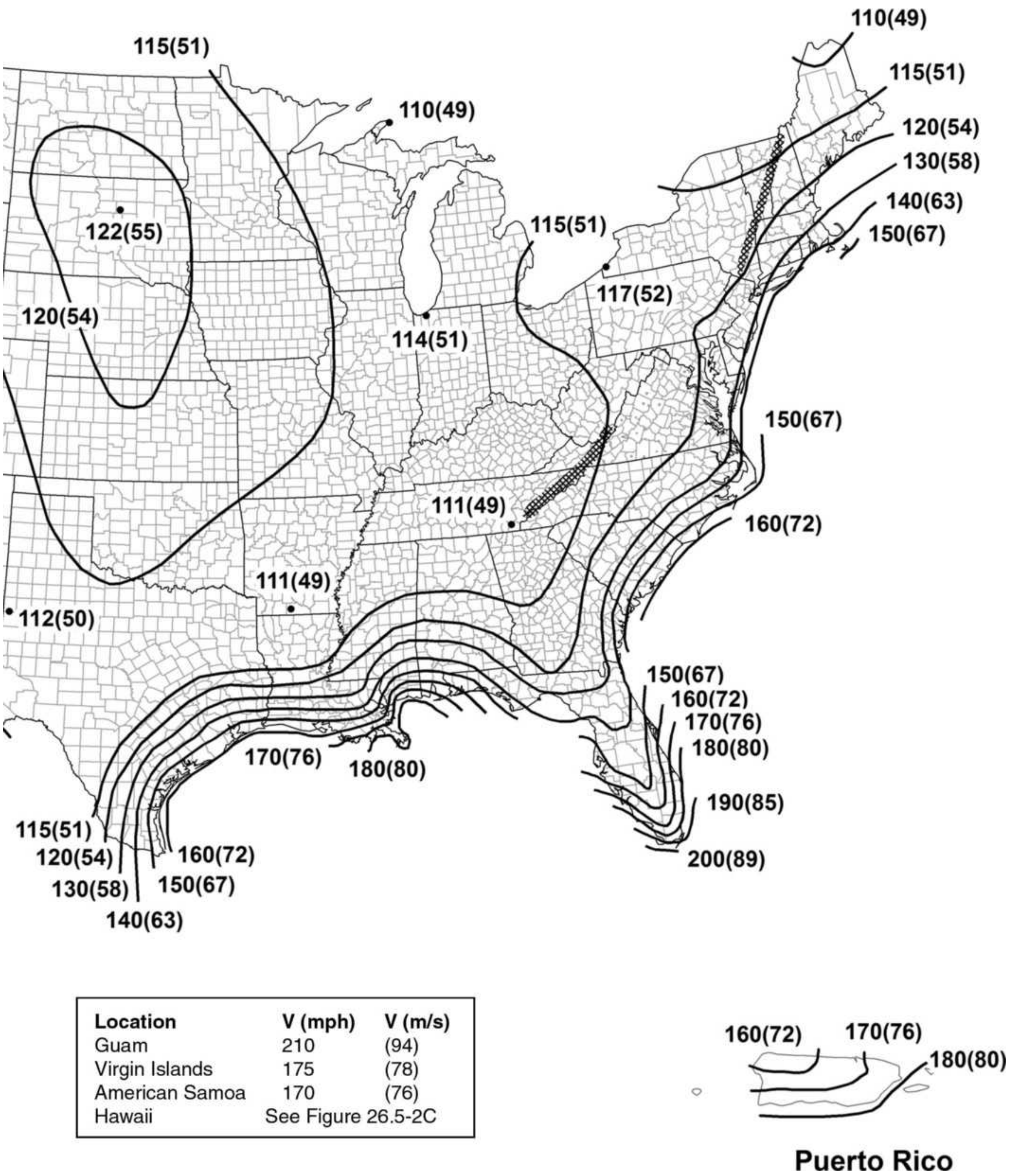
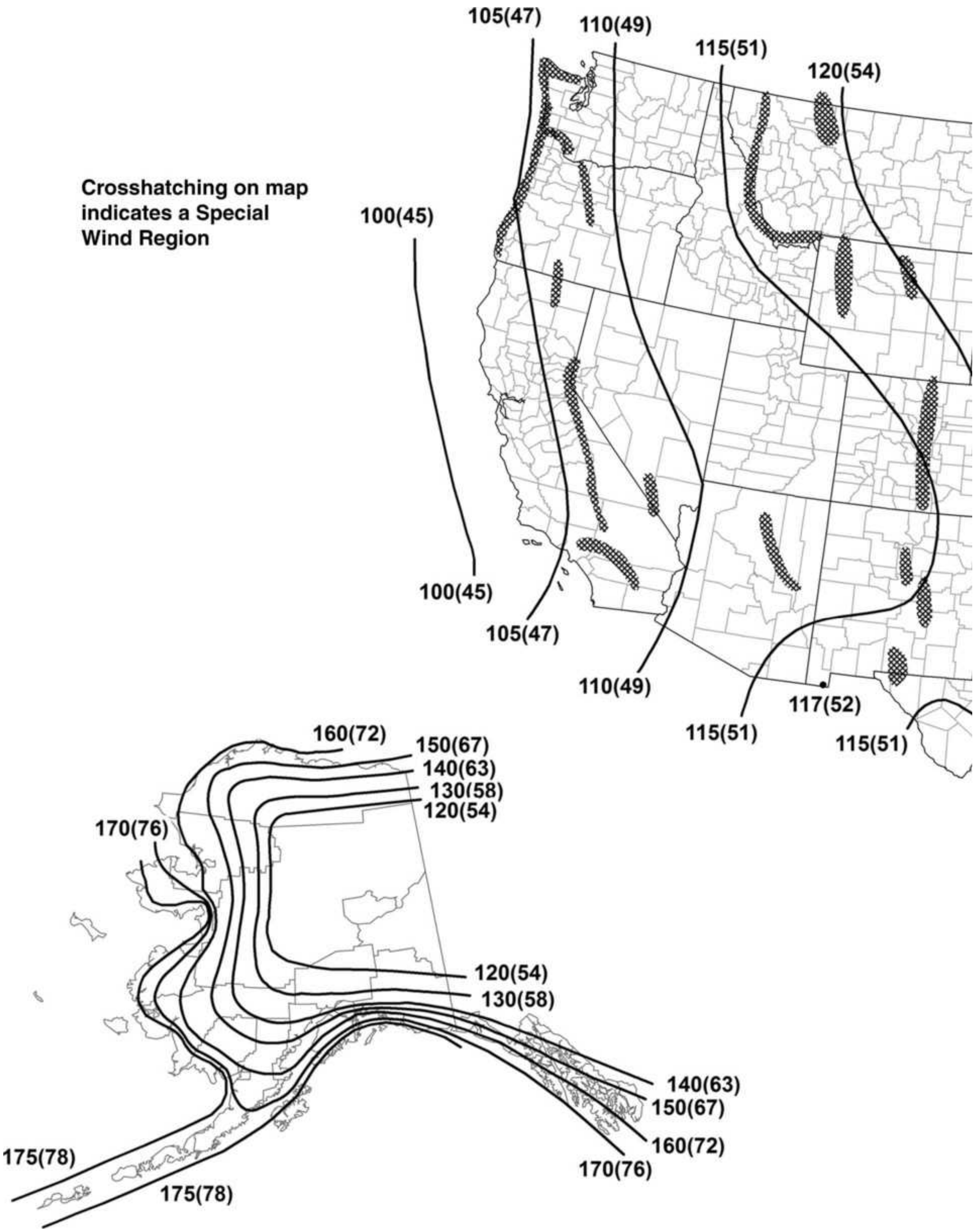


Location	V (mph)	V (m/s)
Guam	210	(94)
Virgin Islands	175	(78)
American Samoa	170	(76)
Hawaii	See Figure 26.5-2C	

FIGURE 26.5-1C (*Continued*). Basic Wind Speeds for Risk Category III Buildings and Other Structures

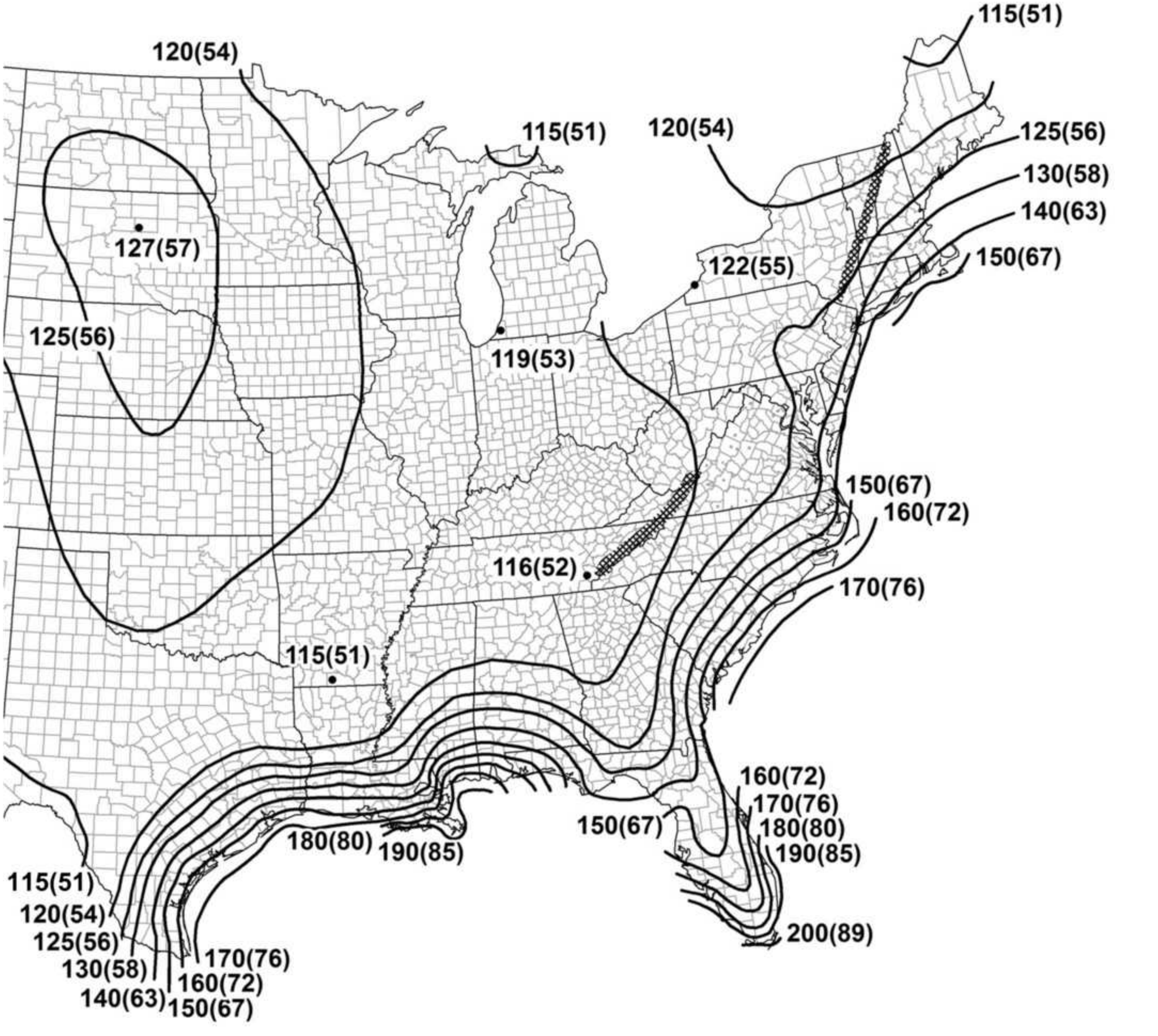


Notes

1. Values are nominal design 3-s gust wind speeds in mi/h (m/s) at 33 ft (10 m) above ground for Exposure Category C.
2. Linear interpolation is permitted between contours. Point values are provided to aid with interpolation.
3. Islands, coastal areas, and land boundaries outside the last contour shall use the last wind speed contour.
4. Mountainous terrain, gorges, ocean promontories, and special wind regions shall be examined for unusual wind conditions.
5. Wind speeds correspond to approximately a 1.6% probability of exceedance in 50 years (Annual Exceedance Probability = 0.00033, MRI = 3,000 years).
6. Location-specific basic wind speeds shall be permitted to be determined using www.atcouncil.org/windspeed.

FIGURE 26.5-1D Basic Wind Speeds for Risk Category IV Buildings and Other Structures

continues



Location	V (mph)	V (m/s)
Guam	180	(80)
Virgin Islands	150	(67)
American Samoa	150	(67)
Hawaii	See Figure 26.5-2D	

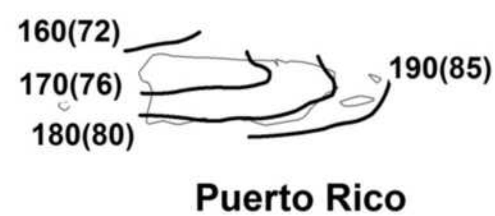
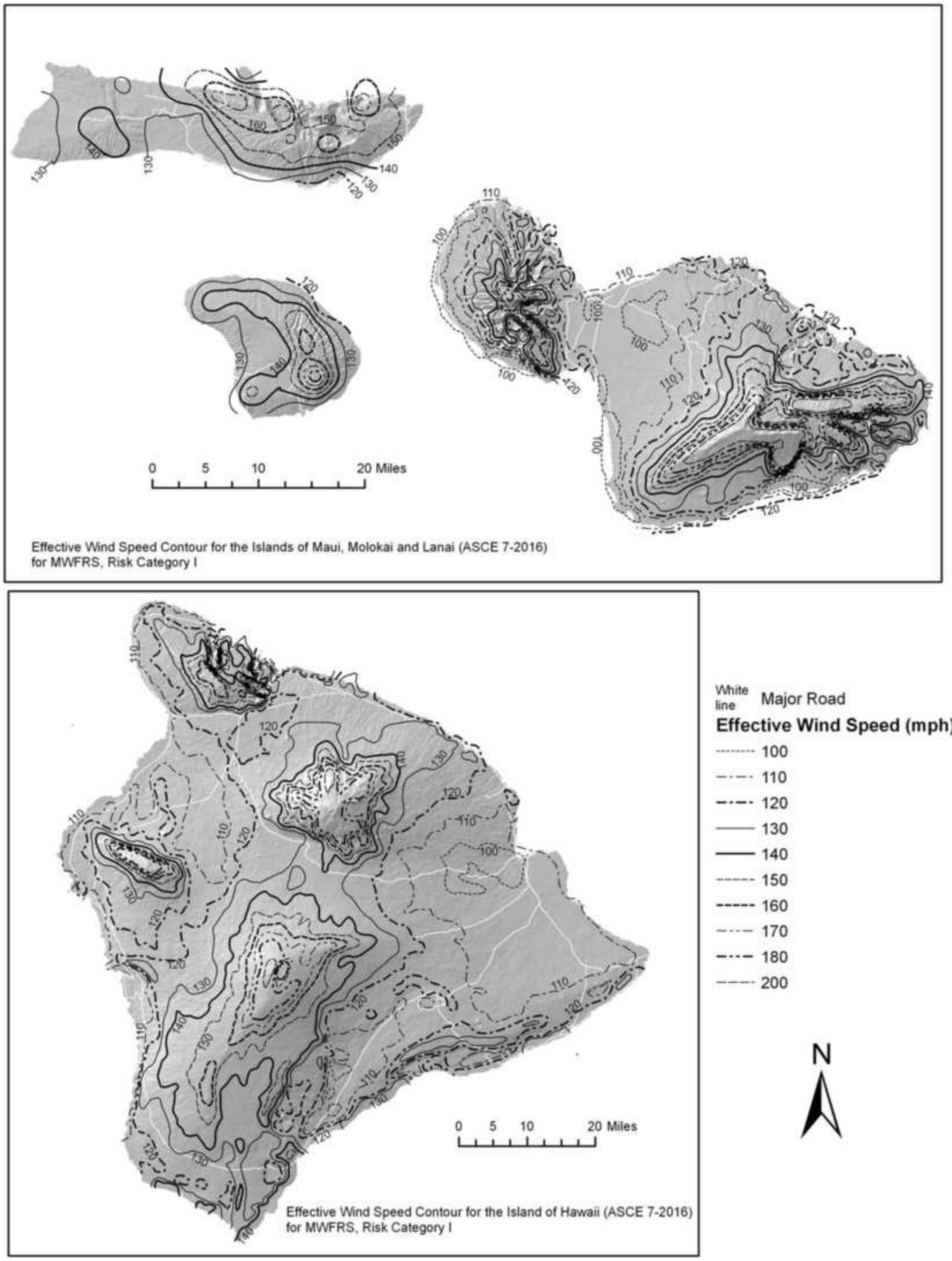


FIGURE 26.5-1D (*Continued*). **Basic Wind Speeds for Risk Category IV Buildings and Other Structures**



Notes

1. Values are nominal design 3-s gust wind speeds in mi/h (m/s) at 33 ft (10 m) above ground for Exposure Category C. Metric conversion: 1 mph = 0.45 m/s.
2. Linear interpolation between contours is permitted.
3. Islands and coastal areas outside the last contour shall use the last wind speed contour of the coastal area.
4. It is permitted to use the standard values of K_{zt} of 1.0 and K_d as given in Table 26.6-1.
5. Ocean promontories and local escarpments shall be examined for unusual wind conditions.
6. Wind speeds correspond to approximately a 15% probability of exceedance in 50 years (Annual Exceedance Probability = 0.00333, MRI = 300 years)

FIGURE 26.5-2A Basic Wind Speeds for Risk Category I Buildings and Other Structures: Hawaii

continues

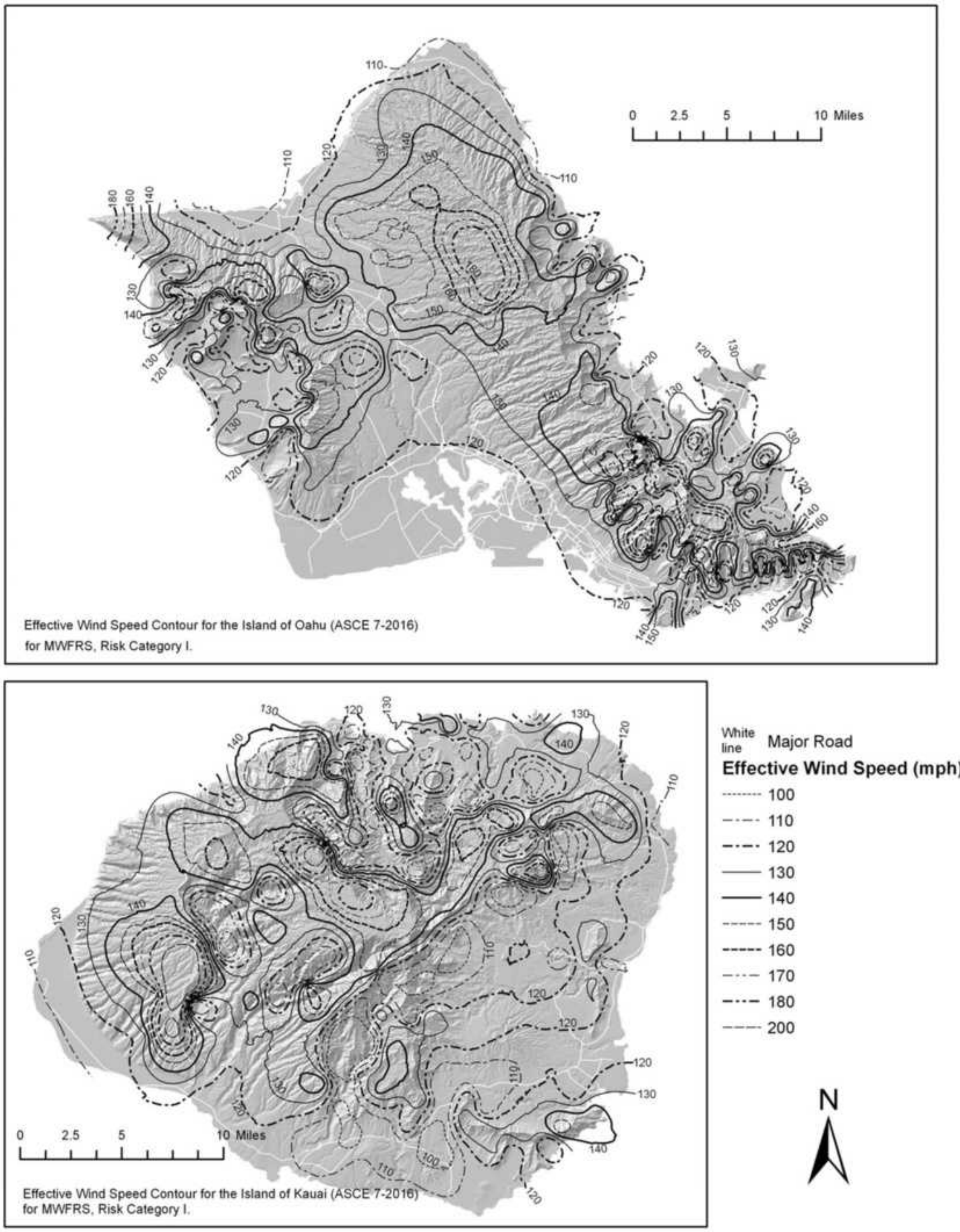
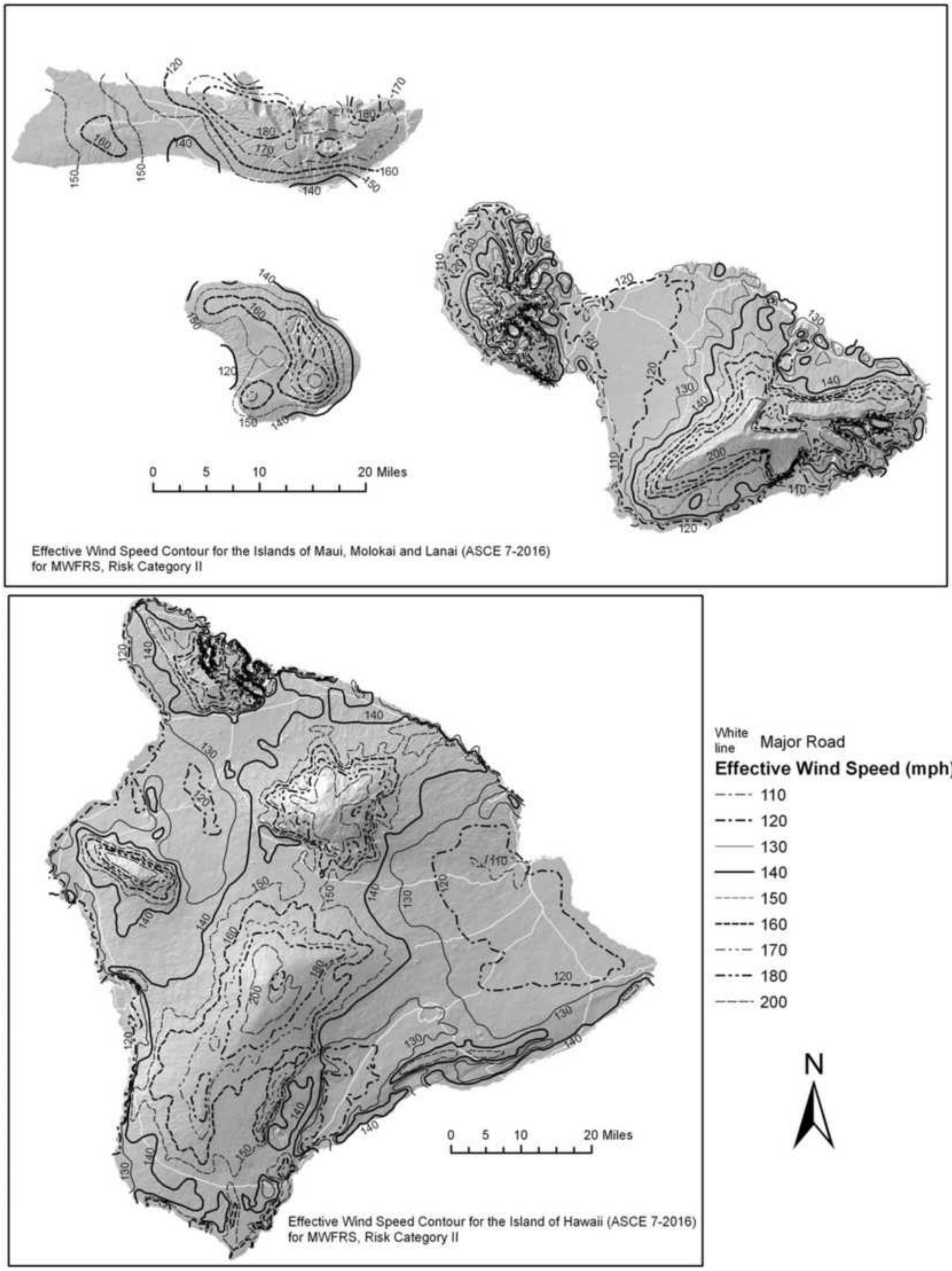


FIGURE 26.5-2A (*Continued*). **Basic Wind Speeds for Risk Category I Buildings and Other Structures: Hawaii**



Notes

1. Values are nominal design 3-s gust wind speeds in mi/h (m/s) at 33 ft (10 m) above ground for Exposure Category C. Metric conversion: 1 mph = 0.45 m/s.
2. Linear interpolation between contours is permitted.
3. Islands and coastal areas outside the last contour shall use the last wind speed contour of the coastal area.
4. It is permitted to use the standard values of K_{zt} of 1.0 and K_d as given in Table 26.6-1.
5. Ocean promontories and local escarpments shall be examined for unusual wind conditions.
6. Wind speeds correspond to approximately a 7% probability of exceedance in 50 years (Annual Exceedance Probability = 0.00143, MRI = 700 years).

FIGURE 26.5-2B Basic Wind Speeds for Risk Category II Buildings and Other Structures: Hawaii

continues

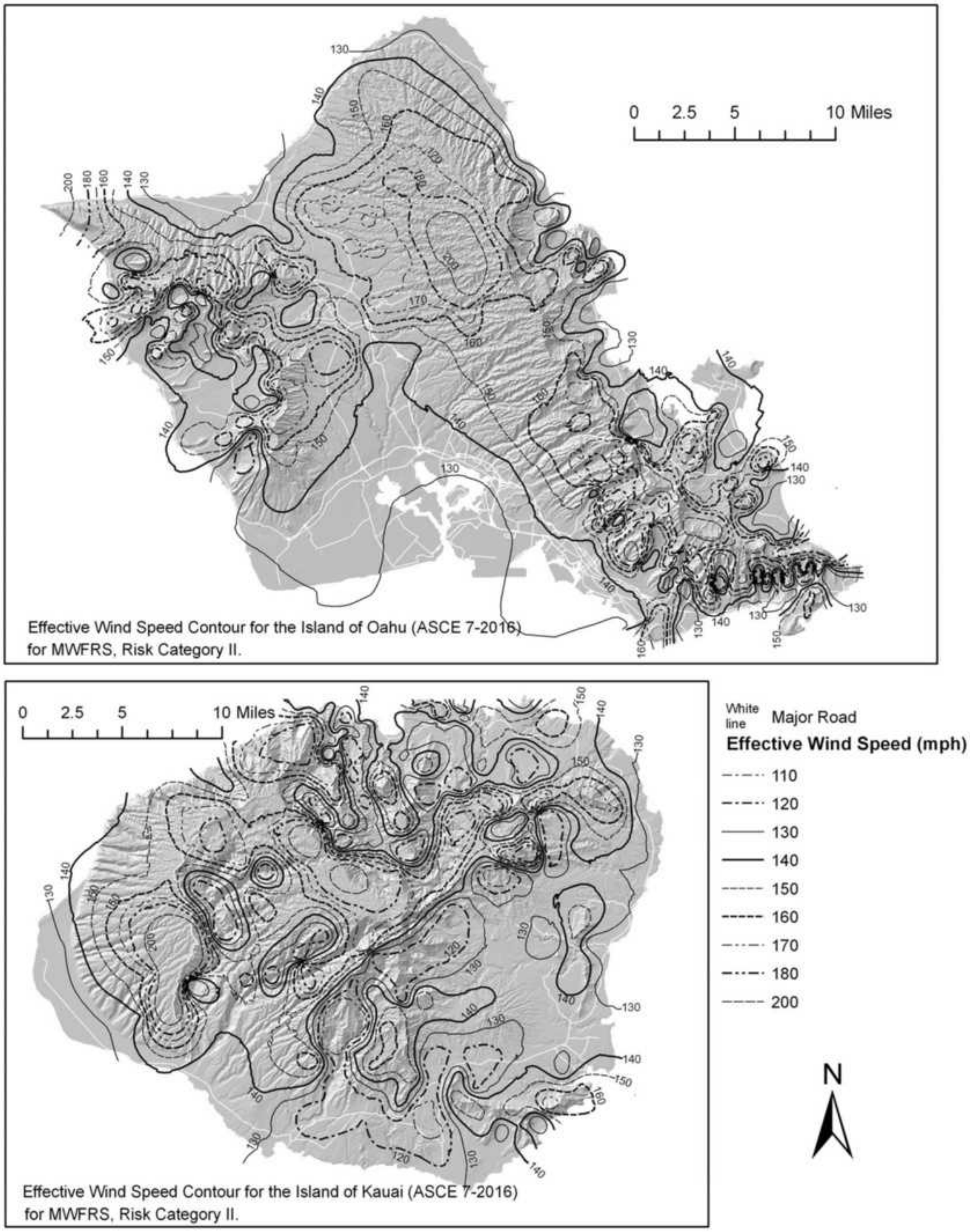
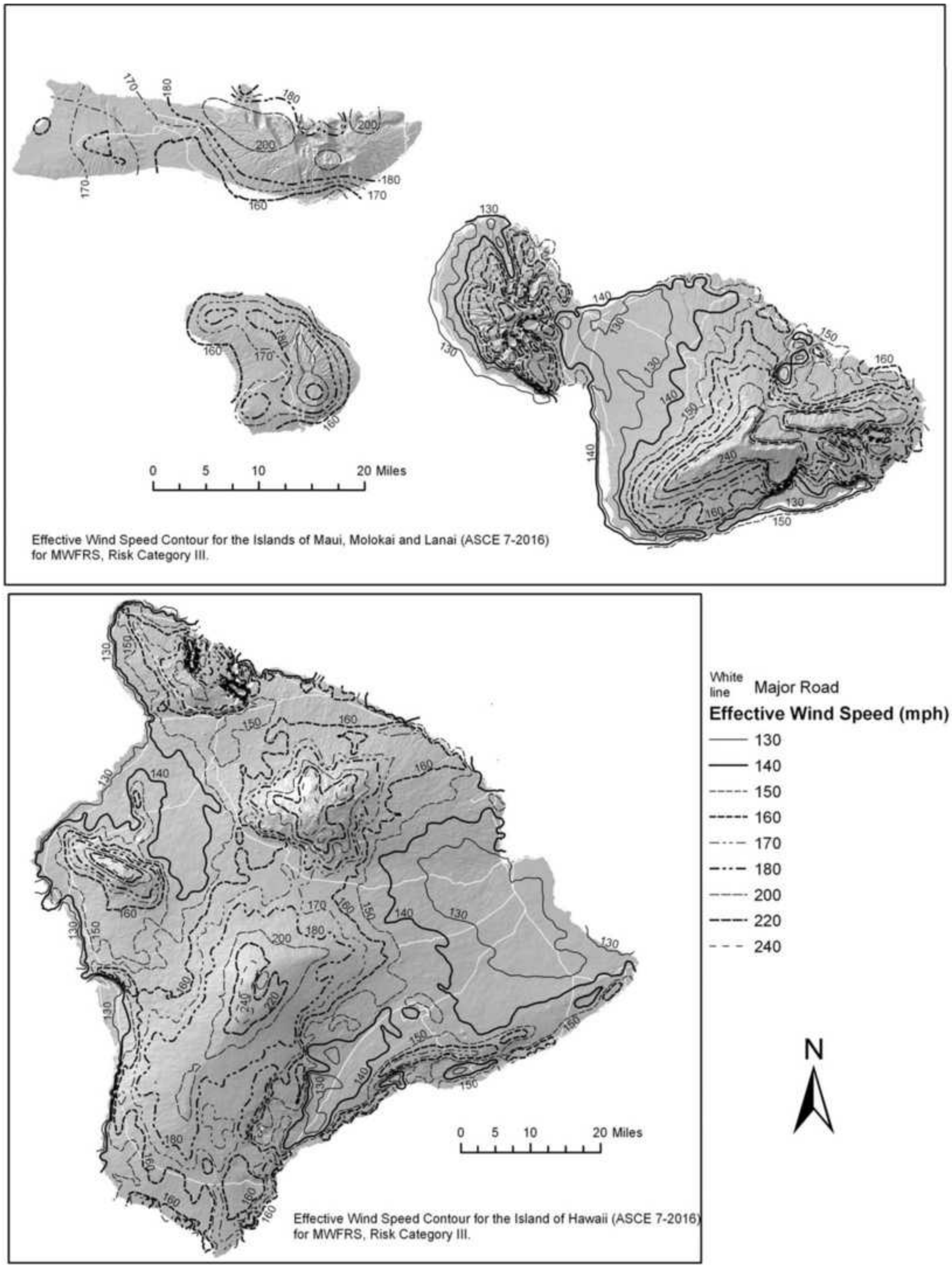


FIGURE 26.5-2B (*Continued*). **Basic Wind Speeds for Risk Category II Buildings and Other Structures: Hawaii**



Notes

1. Values are nominal design 3-s gust wind speeds in mi/h (m/s) at 33 ft (10 m) above ground for Exposure Category C. Metric conversion: 1 mph = 0.45 m/s.
2. Linear interpolation between contours is permitted.
3. Islands and coastal areas outside the last contour shall use the last wind speed contour of the coastal area.
4. It is permitted to use the standard values of K_{zt} of 1.0 and K_d as given in Table 26.6-1.
5. Ocean promontories and local escarpments shall be examined for unusual wind conditions.
6. Wind speeds correspond to approximately a 3% probability of exceedance in 50 years (Annual Exceedance Probability = 0.000588, MRI = 1,700 years).

FIGURE 26.5-2C Basic Wind Speeds for Risk Category III Buildings and Other Structures: Hawaii

continues

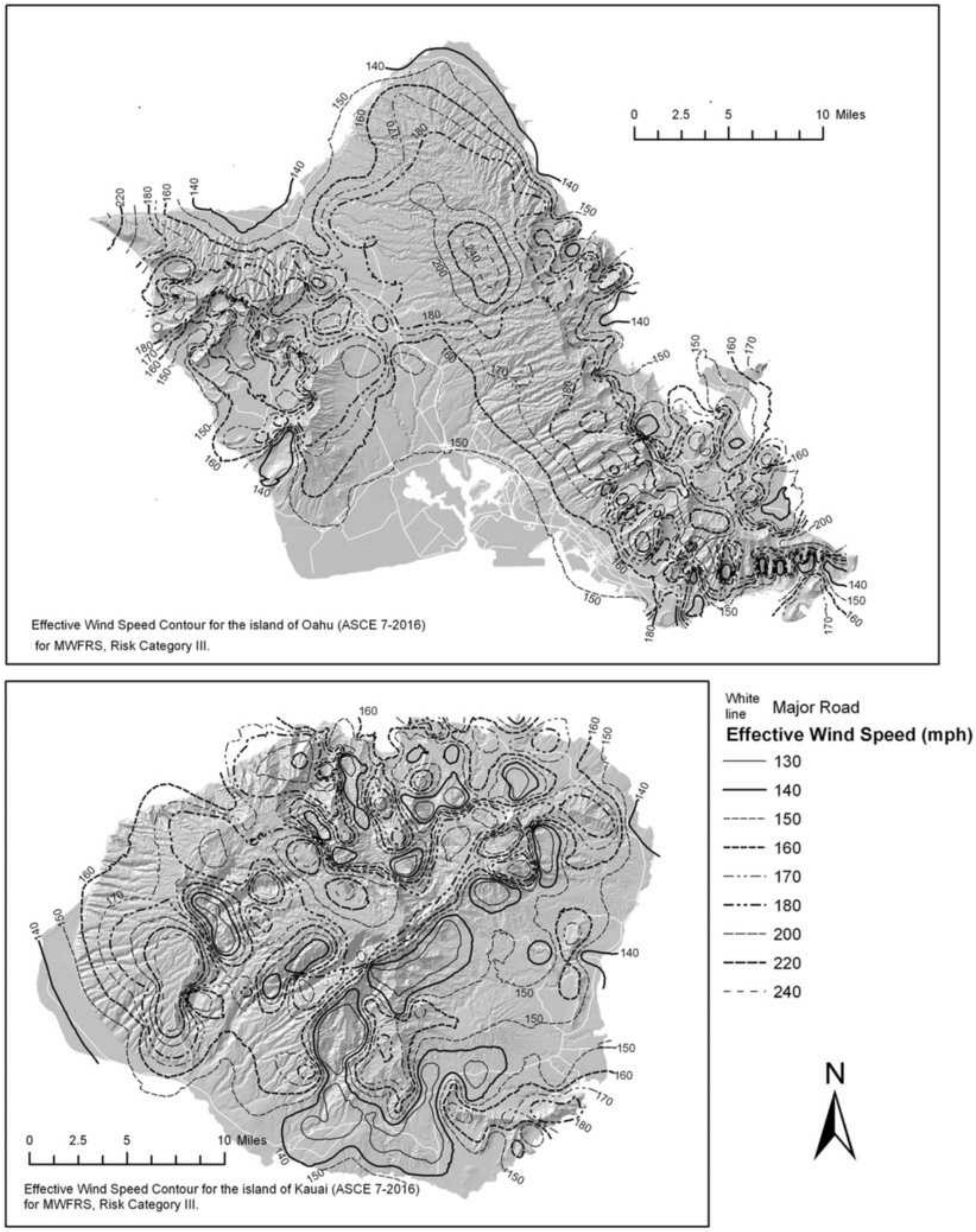
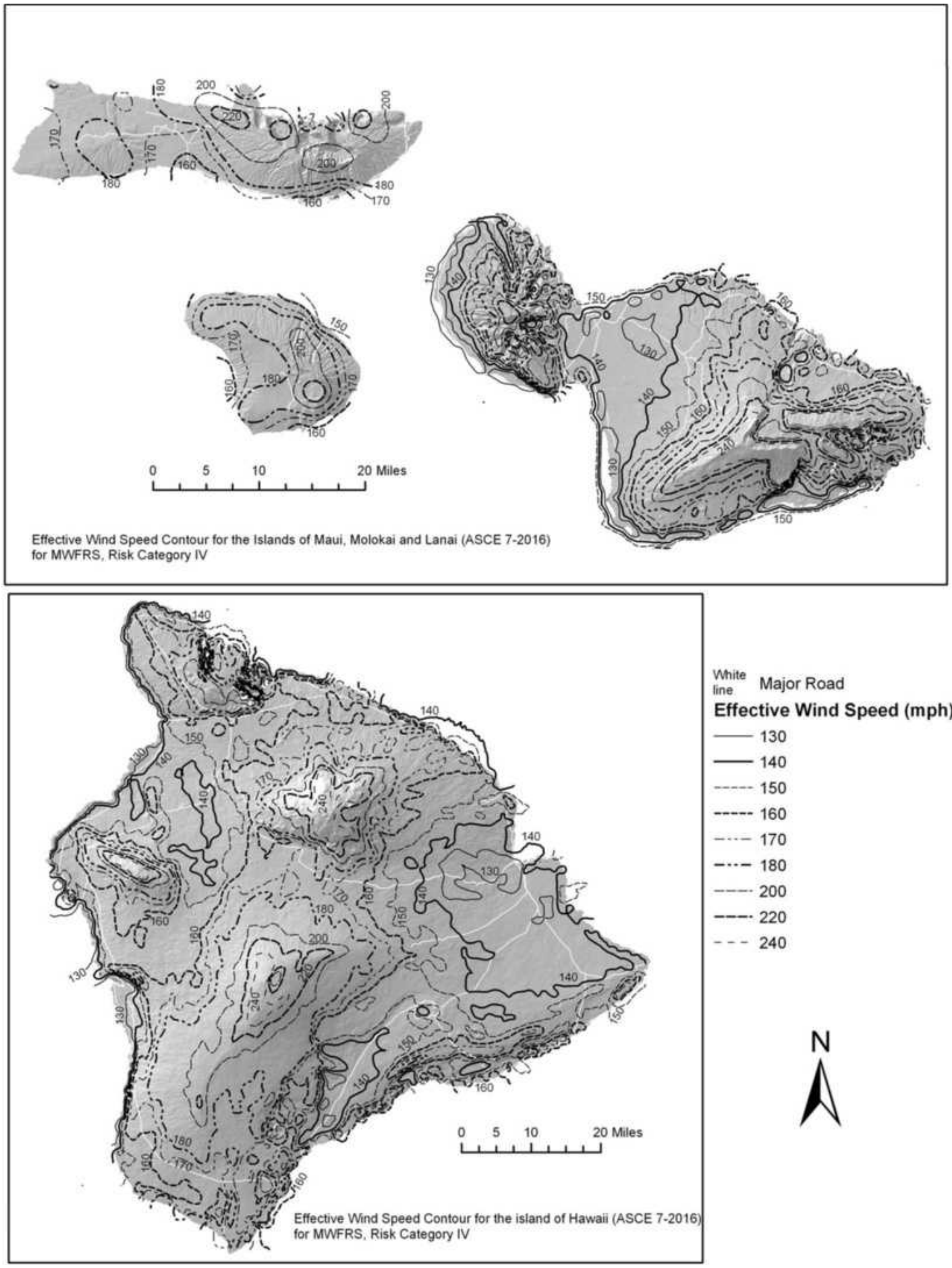


FIGURE 26.5-2C (*Continued*). **Basic Wind Speeds for Risk Category III Buildings and Other Structures: Hawaii**

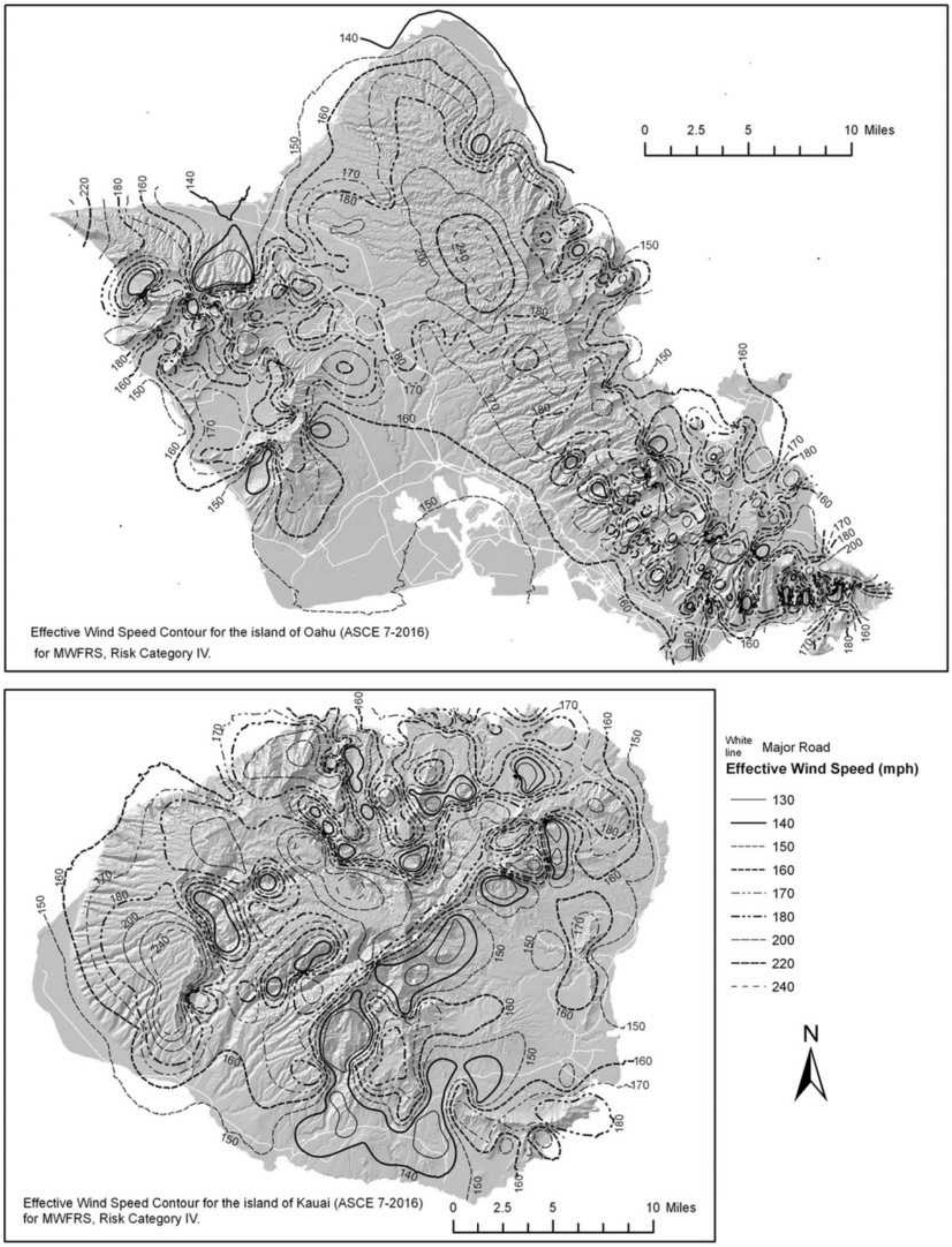


Notes

1. Values are nominal design 3-s gust wind speeds in mi/h (m/s) at 33 ft (10 m) above ground for Exposure Category C. Metric conversion: 1 mph = 0.45 m/s.
2. Linear interpolation between contours is permitted.
3. Islands and coastal areas outside the last contour shall use the last wind speed contour of the coastal area.
4. It is permitted to use the standard values of K_{zt} of 1.0 and K_d as given in Table 26.6-1.
5. Ocean promontories and local escarpments shall be examined for unusual wind conditions.
6. Wind speeds correspond to approximately a 1.7% probability of exceedance in 50 years (Annual Exceedance Probability = 0.000333, MRI = 3,000 years).

FIGURE 26.5-2D Basic Wind Speeds for Risk Category IV Buildings and Other Structures: Hawaii

continues



Effective Wind Speed Contour for the island of Oahu (ASCE 7-2016) for MWFRS, Risk Category IV.

Effective Wind Speed Contour for the island of Kauai (ASCE 7-2016) for MWFRS, Risk Category IV.

FIGURE 26.5-2D (*Continued*). **Basic Wind Speeds for Risk Category IV Buildings and Other Structures: Hawaii**

Table 26.6-1 Wind Directionality Factor, K_d

Structure Type	Directionality Factor K_d
Buildings	
Main Wind Force Resisting System	0.85
Components and Cladding	0.85
Arched Roofs	0.85
Circular Domes	1.0[a]
Chimneys, Tanks, and Similar Structures	
Square	0.90
Hexagonal	0.95
Octagonal	1.0[a]
Round	1.0[a]
Solid Freestanding Walls, Roof Top Equipment, and Solid Freestanding and Attached Signs	0.85
Open Signs and Single-Plane Open Frames	0.85
Trussed Towers	
Triangular, square, or rectangular	0.85
All other cross sections	0.95

[a]Directionality factor $K_d = 0.95$ shall be permitted for round or octagonal structures with nonaxisymmetric structural systems.

26.6 WIND DIRECTIONALITY

The wind directionality factor, K_d, shall be determined from Table 26.6-1 and shall be included in the wind loads calculated in Chapters 27 to 30. The effect of wind directionality in determining wind loads in accordance with Chapter 31 shall be based on a rational analysis of the wind speeds conforming to the requirements of Section 26.5.3 and of Section 31.4.3.

26.7 EXPOSURE

For each wind direction considered, the upwind exposure shall be based on ground surface roughness that is determined from natural topography, vegetation, and constructed facilities.

26.7.1 Wind Directions and Sectors. For each selected wind direction at which the wind loads are to be determined, the exposure of the building or structure shall be determined for the two upwind sectors extending 45° on either side of the selected wind direction. The exposure in these two sectors shall be determined in accordance with Sections 26.7.2 and 26.7.3, and the exposure the use of which would result in the highest wind loads shall be used to represent the winds from that direction.

26.7.2 Surface Roughness Categories. A ground surface roughness within each 45° sector shall be determined for a distance upwind of the site, as defined in Section 26.7.3, from the categories defined in the following text, for the purpose of assigning an exposure category as defined in Section 26.7.3.

Surface Roughness B: Urban and suburban areas, wooded areas, or other terrain with numerous, closely spaced obstructions that have the size of single-family dwellings or larger.

Surface Roughness C: Open terrain with scattered obstructions that have heights generally less than 30 ft (9.1 m). This category includes flat, open country and grasslands.

Surface Roughness D: Flat, unobstructed areas and water surfaces. This category includes smooth mud flats, salt flats, and unbroken ice.

26.7.3 Exposure Categories.

Exposure B: For buildings or other structures with a mean roof height less than or equal to 30 ft (9.1 m), Exposure B shall apply where the ground surface roughness, as defined by Surface Roughness B, prevails in the upwind direction for a distance greater than 1,500 ft (457 m). For buildings or other structures with a mean roof height greater than 30 ft (9.1 m), Exposure B shall apply where Surface Roughness B prevails in the upwind direction for a distance greater than 2,600 ft (792 m) or 20 times the height of the building or structure, whichever is greater.

Exposure C: Exposure C shall apply for all cases where Exposure B or D does not apply.

Exposure D: Exposure D shall apply where the ground surface roughness, as defined by Surface Roughness D, prevails in the upwind direction for a distance greater than 5,000 ft (1,524 m) or 20 times the building or structure height, whichever is greater. Exposure D shall also apply where the ground surface roughness immediately upwind of the site is B or C, and the site is within a distance of 600 ft (183 m) or 20 times the building or structure height, whichever is greater, from an Exposure D condition as defined in the previous sentence.

For a site located in the transition zone between exposure categories, the category resulting in the largest wind forces shall be used.

EXCEPTION: An intermediate exposure between the preceding categories is permitted in a transition zone, provided that it is determined by a rational analysis method defined in the recognized literature.

26.7.4 Exposure Requirements.

26.7.4.1 Directional Procedure (Chapter 27). For each wind direction considered, wind loads for the design of the MWFRS of enclosed and partially enclosed buildings using the Directional Procedure of Chapter 27 shall be based on the exposures as defined in Section 26.7.3. Wind loads for the design of open buildings with monoslope, pitched, or troughed free roofs shall be based on the exposures, as defined in Section 26.7.3, resulting in the highest wind loads for any wind direction at the site.

26.7.4.2 Envelope Procedure (Chapter 28). Wind loads for the design of the MWFRS for all low-rise buildings designed using the Envelope Procedure of Chapter 28 shall be based on the exposure category resulting in the highest wind loads for any wind direction at the site.

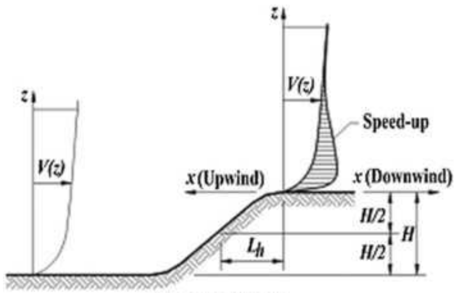
26.7.4.3 Directional Procedure for Building Appurtenances and Other Structures (Chapter 29). Wind loads for the design of building appurtenances (such as rooftop structures and equipment) and other structures (such as solid freestanding walls and freestanding signs, chimneys, tanks, open signs, single-plane open frames, and trussed towers) as specified in Chapter 29 shall be based on the appropriate exposure for each wind direction considered.

26.7.4.4 Components and Cladding (Chapter 30). Design wind pressures for C&C shall be based on the exposure category resulting in the highest wind loads for any wind direction at the site.

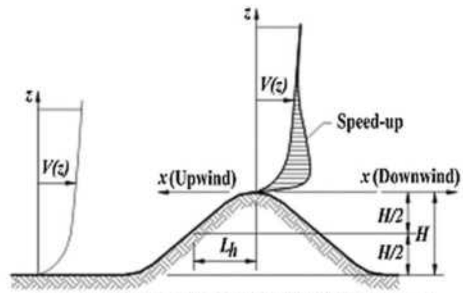
26.8 TOPOGRAPHIC EFFECTS

26.8.1 Wind Speed-Up over Hills, Ridges, and Escarpments. Wind speed-up effects at isolated hills, ridges, and escarpments constituting abrupt changes in the general topography, located in any exposure category, shall be included in the determination of the wind loads when site

Diagrams



ESCARPMENT



2-D RIDGE OR 3-D AXISYMMETRICAL HILL

Topographic Multipliers for Exposure C[a,b,c]

	K_1 Multiplier				K_2 Multiplier			K_3 Multiplier		
H/L_h	2D Ridge	2D Escarpment	3D Axisymmetrical Hill	x/L_h	2D Escarpment	All Other Cases	z/L_h	2D Ridge	2D Escarpment	3D Axisymmetrical Hill
0.20	0.29	0.17	0.21	0.00	1.00	1.00	0.00	1.00	1.00	1.00
0.25	0.36	0.21	0.26	0.50	0.88	0.67	0.10	0.74	0.78	0.67
0.30	0.43	0.26	0.32	1.00	0.75	0.33	0.20	0.55	0.61	0.45
0.35	0.51	0.30	0.37	1.50	0.63	0.00	0.30	0.41	0.47	0.30
0.40	0.58	0.34	0.42	2.00	0.50	0.00	0.40	0.30	0.37	0.20
0.45	0.65	0.38	0.47	2.50	0.38	0.00	0.50	0.22	0.29	0.14
0.50	0.72	0.43	0.53	3.00	0.25	0.00	0.60	0.17	0.22	0.09
				3.50	0.13	0.00	0.70	0.12	0.17	0.06
				4.00	0.00	0.00	0.80	0.09	0.14	0.04
							0.90	0.07	0.11	0.03
							1.00	0.05	0.08	0.02
							0.50	0.01	0.02	0.00
							2.00	0.00	0.00	0.00

[a]For values of H/L_h, x/L_h, and z/L_h other than those shown, linear interpolation is permitted.
[b]For $H/L_h > 0.5$, assume that $H/L_h = 0.5$ for evaluating K_1 and substitute $2H$ for L_h for evaluating K_2 and K_3.
[c]Multipliers are based on the assumption that wind approaches the hill or escarpment along the direction of maximum slope.

Notation

H = Height of hill or escarpment relative to the upwind terrain, in ft (m).
K_1 = Factor to account for shape of topographic feature and maximum speed-up effect.
K_2 = Factor to account for reduction in speed-up with distance upwind or downwind of crest.
K_3 = Factor to account for reduction in speed-up with height above local terrain.
L_h = Distance upwind of crest to where the difference in ground elevation is half the height of hill or escarpment, in ft (m).
x = Distance (upwind or downwind) from the crest to the site of the building or other structure, in ft (m).
z = Height above ground surface at the site of the building or other structure, in ft (m).
μ = Horizontal attenuation factor.
γ = Height attenuation factor.

Equations

$K_{zt} = (1 + K_1 K_2 K_3)^2$
K_1 = determined from table below
$K_2 = (1 - |x|/\mu L_h)$
$K_3 = e^{-\gamma z/L_h}$

Parameters for Speed-Up over Hills and Escarpments

	$K_1/(H/L_h)$				μ	
	Exposure					
Hill Shape	B	C	D	γ	Upwind of Crest	Downwind of Crest
2D ridges (or valleys with negative H in $K_1/(H/L_h)$	1.30	1.45	1.55	3	1.5	1.5
2D escarpments	0.75	0.85	0.95	2.5	1.5	4
3D axisymmetrical hill	0.95	1.05	1.15	4	1.5	1.5

FIGURE 26.8-1 Topographic Factor, K_{zt}

conditions and locations of buildings and other structures meet all of the following conditions:

1. The hill, ridge, or escarpment is isolated and unobstructed upwind by other similar topographic features of comparable height for 100 times the height of the topographic feature ($100H$) or 2 mi (3.22 km), whichever is less. This distance shall be measured horizontally from the point at which the height H of the hill, ridge, or escarpment is determined.
2. The hill, ridge, or escarpment protrudes above the height of upwind terrain features within a 2-mi (3.22-km) radius in any quadrant by a factor of 2 or more.
3. The building or other structure is located as shown in Fig. 26.8-1 in the upper one-half of a hill or ridge or near the crest of an escarpment.
4. $H/L_h \geq 0.2$.
5. H is greater than or equal to 15 ft (4.5 m) for Exposure C and D and 60 ft (18 m) for Exposure B.

26.8.2 Topographic Factor. The wind speed-up effect shall be included in the calculation of design wind loads by using the factor K_{zt}:

$$K_{zt} = (1 + K_1 K_2 K_3)^2 \qquad (26.8\text{-}1)$$

where K_1, K_2, and K_3 are given in Fig. 26.8-1.

If site conditions and locations of buildings and other structures do not meet all the conditions specified in Section 26.8.1, then $K_{zt} = 1.0$.

26.9 GROUND ELEVATION FACTOR

The ground elevation factor to adjust for air density, K_e, shall be determined in accordance with Table 26.9-1. It is permitted to take $K_e = 1$ for all elevations.

26.10 VELOCITY PRESSURE

26.10.1 Velocity Pressure Exposure Coefficient. Based on the exposure category determined in Section 26.7.3, a velocity pressure exposure coefficient, K_z or K_h, as applicable, shall be determined from Table 26.10-1. For a site located in a transition zone between exposure categories that is near to a change in ground surface roughness, intermediate values of K_z or K_h, between those shown in Table 26.10-1 are permitted provided that they are determined by a rational analysis method defined in the recognized literature.

Table 26.9-1 Ground Elevation Factor, K_e

Ground Elevation above Sea Level		Ground Elevation Factor
ft	m	K_e
<0	<0	See note 2
0	0	1.00
1,000	305	0.96
2,000	610	0.93
3,000	914	0.90
4,000	1,219	0.86
5,000	1,524	0.83
6,000	1,829	0.80
>6,000	>1,829	See note 2

Notes

1. The conservative approximation $K_e = 1.00$ is permitted in all cases.
2. The factor K_e shall be determined from the above table using interpolation or from the following formula for all elevations:
   $K_e = e^{-0.0000362 z_g}$ (z_g = ground elevation above sea level in ft).
   $K_e = e^{-0.000119 z_g}$ (z_g = ground elevation above sea level in m).
3. K_e is permitted to be take as 1.00 in all cases.

Table 26.10-1 Velocity Pressure Exposure Coefficients, K_h and K_z

Height above Ground Level, z		Exposure		
ft	m	B	C	D
0–15	0–4.6	0.57 (0.70)[a]	0.85	1.03
20	6.1	0.62 (0.70)[a]	0.90	1.08
25	7.6	0.66 (0.70)[a]	0.94	1.12
30	9.1	0.70	0.98	1.16
40	12.2	0.76	1.04	1.22
50	15.2	0.81	1.09	1.27
60	18.0	0.85	1.13	1.31
70	21.3	0.89	1.17	1.34
80	24.4	0.93	1.21	1.38
90	27.4	0.96	1.24	1.40
100	30.5	0.99	1.26	1.43
120	36.6	1.04	1.31	1.48
140	42.7	1.09	1.36	1.52
160	48.8	1.13	1.39	1.55
180	54.9	1.17	1.43	1.58
200	61.0	1.20	1.46	1.61
250	76.2	1.28	1.53	1.68
300	91.4	1.35	1.59	1.73
350	106.7	1.41	1.64	1.78
400	121.9	1.47	1.69	1.82
450	137.2	1.52	1.73	1.86
500	152.4	1.56	1.77	1.89

[a] Use 0.70 in Chapter 28, Exposure B, when $z < 30$ ft (9.1 m).

Notes

1. The velocity pressure exposure coefficient K_z may be determined from the following formula:
   For 15 ft (4.6 m) $\leq z \leq z_g$ $\quad K_z = 2.01(z/z_g)^{2/\alpha}$
   For $z < 15$ ft (4.6 m) $\quad K_z = 2.01(15/z_g)^{2/\alpha}$
2. α and z_g are tabulated in Table 26.11-1.
3. Linear interpolation for intermediate values of height z is acceptable.
4. Exposure categories are defined in Section 26.7.

26.10.2 Velocity Pressure. Velocity pressure, q_z, evaluated at height z above ground shall be calculated by the following equation:

$$q_z = 0.00256 K_z K_{zt} K_d K_e V^2 \text{ (lb/ft}^2\text{); } V \text{ in mi/h} \qquad (26.10\text{-}1)$$

$$q_z = 0.613 K_z K_{zt} K_d K_e V^2 \text{ (N/m}^2\text{); } V \text{ in m/s} \qquad (26.10\text{-}1.\text{si})$$

where

K_z = velocity pressure exposure coefficient, see Section 26.10.1.
K_{zt} = topographic factor, see Section 26.8.2.
K_d = wind directionality factor, see Section 26.6.
K_e = ground elevation factor, see Section 26.9.
V = basic wind speed, see Section 26.5.
q_z = velocity pressure at height z.

The velocity pressure at mean roof height is computed as $q_h = q_z$ evaluated from Eq. (26.10-1) using K_z at mean roof height h.

The basic wind speed, V, used in determination of design wind loads on rooftop structures, rooftop equipment, and other

Table 26.11-1 Terrain Exposure Constants

Customary Units										
Exposure	α	z_g (ft)	$\hat{\alpha}$	$\hat{b}$	$\bar{\alpha}$	$\bar{b}$	c	ℓ (ft)	$\bar{\epsilon}$	$z_{\min}$ (ft)[a]
B	7.0	1,200	1/70	0.84	1/4.0	0.45	0.30	320	1/3.0	30
C	9.5	900	1/9.5	1.00	1/6.5	0.65	0.20	500	1/5.0	15
D	11.5	700	1/11.5	1.07	1/9.0	0.80	0.15	650	1/8.0	7
S.I. Units										
Exposure	α	z_g (m)	$\hat{\alpha}$	$\hat{b}$	$\bar{\alpha}$	$\bar{b}$	c	ℓ (m)	$\bar{\epsilon}$	$z_{\min}$ (m)[a]
B	7.0	365.76	1/7	0.84	1/4.0	0.45	0.30	97.54	1/3.0	9.14
C	9.5	274.32	1/9.5	1.00	1/6.5	0.65	0.20	152.40	1/5.0	4.57
D	11.5	213.36	1/11.5	1.07	1/9.0	0.80	0.15	198.12	1/8.0	2.13

[a] $z_{\min}$ = minimum height used to ensure that the equivalent height $\bar{z}$ is the greater of $0.6h$ or $z_{\min}$. For buildings or other structures with $h \le z_{\min}$, $\bar{z}$ shall be taken as $z_{\min}$.

building appurtenances shall consider the Risk Category equal to the greater of the following:

1. Risk Category for the building on which the equipment or appurtenance is located or
2. Risk Category for any facility to which the equipment or appurtenance provides a necessary service.

26.11 GUST EFFECTS

26.11.1 Gust-Effect Factor. The gust-effect factor for a rigid building or other structure is permitted to be taken as 0.85.

26.11.2 Frequency Determination. To determine whether a building or other structure is rigid or flexible as defined in Section 26.2, the fundamental natural frequency, n_1, shall be established using the structural properties and deformational characteristics of the resisting elements in a properly substantiated analysis. Low-rise buildings, as defined in Section 26.2, are permitted to be considered rigid.

26.11.2.1 Limitations for Approximate Natural Frequency. As an alternative to performing an analysis to determine n_1, the approximate building natural frequency, n_a, shall be permitted to be calculated in accordance with Section 26.11.3 for structural steel, concrete, or masonry buildings meeting the following requirements:

1. The building height is less than or equal to 300 ft (91 m), and
2. The building height is less than 4 times its effective length, L_{eff}.

The effective length, L_{eff}, in the direction under consideration shall be determined from the following equation:

$$L_{\text{eff}} = \frac{\sum_{i=1}^{n} h_i L_i}{\sum_{i=1}^{n} h_i} \qquad (26.11\text{-}1)$$

The summations are over the height of the building where

h_i = height above grade of level i; and
L_i = building length at level i parallel to the wind direction.

26.11.3 Approximate Natural Frequency. The approximate lower bound natural frequency (n_a), in hertz, of concrete or structural steel buildings meeting the conditions of Section 26.11.2.1 is permitted to be determined from one of the following equations:

For structural steel moment-resisting frame buildings,

$$n_a = 22.2/h^{0.8} \qquad (26.11\text{-}2)$$

For concrete moment-resisting frame buildings,

$$n_a = 43.5/h^{0.9} \qquad (26.11\text{-}3)$$

For structural steel and concrete buildings with other lateral-force-resisting systems,

$$n_a = 75/h \qquad (26.11\text{-}4)$$

For concrete or masonry shear wall buildings, it is also permitted to use

$$n_a = 385(C_w)^{0.5}/h \qquad (26.11\text{-}5)$$

where

$$C_w = \frac{100}{A_B}\sum_{i=1}^{n}\left(\frac{h}{h_i}\right)^2 \frac{A_i}{\left[1+0.83\left(\frac{h_i}{D_i}\right)^2\right]}$$

where

h = mean roof height, ft (m).
n = number of shear walls in the building effective in resisting lateral forces in the direction under consideration.
A_B = base area of the building, ft^2 (m^2).
A_i = horizontal cross-sectional area of shear wall i, ft^2 (m^2).
D_i = length of shear wall i, ft (m).
h_i = height of shear wall i, ft (m).

26.11.4 Rigid Buildings or Other Structures. For rigid buildings or other structures as defined in Section 26.2, the gust-effect factor shall be taken as 0.85 or calculated by this formula:

$$G = 0.925\left(\frac{1+0.7 g_Q I_{\bar{z}} Q}{1+0.7 g_v I_{\bar{z}}}\right) \qquad (26.11\text{-}6)$$

$$I_{\bar{z}} = c\left(\frac{33}{\bar{z}}\right)^{1/6} \qquad (26.11\text{-}7)$$

$$I_{\bar{z}} = c\left(\frac{10}{\bar{z}}\right)^{1/6} \qquad (26.11\text{-}7.\text{si})$$

where $I_{\bar{z}}$ = intensity of turbulence at height $\bar{z}$, where $\bar{z}$ is the equivalent height of the building or structure defined as $0.6h$, but not less than $z_{\min}$ for all building or structure heights h. $z_{\min}$ and c are listed for each exposure in Table 26.11-1; g_Q and g_v shall be taken as 3.4. The background response Q is given by

$$Q = \sqrt{\frac{1}{1 + 0.63\left(\frac{B+h}{L_{\bar{z}}}\right)^{0.63}}} \qquad (26.11\text{-}8)$$

where B and h are defined in Section 26.3; and $L_{\bar{z}}$ = integral length scale of turbulence at the equivalent height given by

$$L_{\bar{z}} = \ell\left(\frac{\bar{z}}{33}\right)^{\bar{\varepsilon}} \qquad (26.11\text{-}9)$$

$$L_{\bar{z}} = \ell\left(\frac{\bar{z}}{10}\right)^{\bar{\varepsilon}} \qquad (26.11\text{-}9.\text{si})$$

in which ℓ and $\bar{\varepsilon}$ = constants listed in Table 26.11-1.

26.11.5 Flexible or Dynamically Sensitive Buildings or Other Structures. For flexible or dynamically sensitive buildings or other structures as defined in Section 26.2, the gust-effect factor shall be calculated by

$$G_f = 0.925\left(\frac{1 + 1.7I_{\bar{z}}\sqrt{g_Q^2 Q^2 + g_R^2 R^2}}{1 + 1.7g_v I_{\bar{z}}}\right) \qquad (26.11\text{-}10)$$

g_Q and g_v shall be taken as 3.4, and g_R is given by

$$g_R = \sqrt{2\ln(3{,}600n_1)} + \frac{0.577}{\sqrt{2\ \ln(3{,}600n_1)}} \qquad (26.11\text{-}11)$$

R, the resonant response factor, is given by

$$R = \sqrt{\frac{1}{\beta}R_n R_h R_B(0.53 + 0.47R_L)} \qquad (26.11\text{-}12)$$

$$R_n = \frac{7.47N_1}{(1 + 10.3N_1)^{5/3}} \qquad (26.11\text{-}13)$$

$$N_1 = \frac{n_1 L_{\bar{z}}}{\bar{V}_{\bar{z}}} \qquad (26.11\text{-}14)$$

$$R_\ell = \frac{1}{\eta} - \frac{1}{2\eta^2}(1 - e^{-2\eta}) \quad \text{for } \eta > 0 \qquad (26.11\text{-}15a)$$

$$R_l = 1 \quad \text{for } \eta = 0 \qquad (26.11\text{-}15b)$$

where the subscript ℓ in Eqs. (26.11-15a) and (26.11-15b) shall be taken as h, B, and L, respectively, where h, B, and L are defined in Section 26.3, and

n_1 = fundamental natural frequency.
$R_\ell = R_h$ setting $\eta = 4.6n_1 h/\bar{V}_{\bar{z}}$.
$R_\ell = R_B$ setting $\eta = 4.6n_1 B/\bar{V}_{\bar{z}}$.
$R_\ell = R_L$ setting $\eta = 15.4n_1 L/\bar{V}_{\bar{z}}$;
β = damping ratio, percent of critical (i.e., for 2% use 0.02 in the equation).
$\bar{V}_{\bar{z}}$ = mean hourly wind speed (ft/s) (m/s) at height $\bar{z}$ determined from Eq. (26.11-16):

$$\bar{V}_{\bar{z}} = \bar{b}\left(\frac{\bar{z}}{33}\right)^{\bar{\alpha}}\left(\frac{88}{60}\right)V \qquad (26.11\text{-}16)$$

$$\bar{V}_{\bar{z}} = \bar{b}\left(\frac{\bar{z}}{10}\right)^{\bar{\alpha}}V \qquad (26.11\text{-}16.\text{si})$$

where $\bar{b}$ and $\bar{\alpha}$ are constants listed in Table 26.9-1; and V is the basic wind speed in mi/h (m/s).

26.11.6 Rational Analysis. In lieu of the procedure defined in Sections 26.11.4 and 26.11.5, determination of the gust-effect factor by any rational analysis defined in the recognized literature is permitted.

26.11.7 Limitations. Where combined gust-effect factors and pressure coefficients (GC_p), (GC_{pi}), and (GC_{pf}) are given in figures and tables, the gust-effect factor shall not be determined separately.

26.12 ENCLOSURE CLASSIFICATION

26.12.1 General. For the purpose of determining internal pressure coefficients, all buildings shall be classified as enclosed, partially enclosed, partially open, or open as defined in Section 26.2.

26.12.2 Openings. A determination shall be made of the amount of openings in the building envelope for use in determining the enclosure classification. To make this determination, each building wall shall be assumed as the windward wall for consideration of the amount of openings present with respect to the remaining building envelope.

26.12.3 Protection of Glazed Openings. Glazed openings in Risk Category II, III, or IV buildings located in hurricane-prone regions shall be protected as specified in this section.

26.12.3.1 Wind-Borne Debris Regions. Glazed openings shall be protected in accordance with Section 26.12.3.2 in the following locations:

1. Within 1 mi (1.6 km) of the coastal mean high water line where the basic wind speed is equal to or greater than 130 mi/h (58 m/s), or
2. In areas where the basic wind speed is equal to or greater than 140 mi/h (63 m/s).

For Risk Category II buildings and other structures and Risk Category III buildings and other structures, except health-care facilities, the wind-borne debris region shall be based on Figs. 26.5-1B and 26.5-2B. For Risk Category III health-care facilities, the wind-borne debris region shall be based on Figs. 26.5-1C and 26.5-2C. For Risk Category IV buildings and structures, the wind-borne debris region shall be based on Figs. 26.5-1D and 26.5-2D. Risk Categories shall be determined in accordance with Section 1.5.

EXCEPTION: Glazing located more than 60 ft (18.3 m) above the ground and more than 30 ft (9.2 m) above aggregate-surfaced roofs, including roofs with gravel or stone ballast, located within 1,500 ft (458 m) of the building shall be permitted to be unprotected.

26.12.3.2 Protection Requirements for Glazed Openings. Glazing in buildings requiring protection shall be protected with an impact-protective system or shall be impact-resistant glazing.

Impact-protective systems and impact-resistant glazing shall be subjected to missile test and cyclic pressure differential tests in accordance with ASTM E1996 as applicable. Testing to demonstrate compliance with ASTM E1996 shall be in accordance with ASTM E1886. Impact-resistant glazing and impact-protective systems shall comply with the pass/fail criteria of Section 7 of ASTM E1996 based on the missile required by Table 3 or Table 4 of ASTM E1996. Glazing in sectional garage doors and rolling doors shall be subjected to missile tests and cyclic pressure differential tests in accordance with ANSI/DASMA 115 as applicable.

Table 26.13-1 Main Wind Force Resisting System and Components and Cladding (All Heights): Internal Pressure Coefficient, (GC_{pi}), for Enclosed, Partially Enclosed, Partially Open, and Open Buildings (Walls and Roof)

Enclosure Classification	Criteria for Enclosure Classification	Internal Pressure	Internal Pressure Coefficient, (GC_{pi})
Enclosed buildings	A_o is less than the smaller of $0.01A_g$ or 4 sq ft (0.37 m) and $A_{oi}/A_{gi} \leq 0.2$	Moderate	+0.18 −0.18
Partially enclosed buildings	$A_o > 1.1A_{oi}$ and $A_o >$ the lesser of $0.01A_g$ or 4 sq ft (0.37 m) and $A_{oi}/A_{gi} \leq 0.2$	High	+0.55 −0.55
Partially open buildings	A building that does not comply with Enclosed, Partially Enclosed, or Open classifications	Moderate	+0.18 −0.18
Open buildings	Each wall is at least 80% open	Negligible	0.00

Notes

1. Plus and minus signs signify pressures acting toward and away from the internal surfaces, respectively.
2. Values of (GC_{pi}) shall be used with q_z or q_h as specified.
3. Two cases shall be considered to determine the critical load requirements for the appropriate condition:
   a. A positive value of (GC_{pi}) applied to all internal surfaces, or
   b. A negative value of (GC_{pi}) applied to all internal surfaces.

EXCEPTION: Other testing methods and/or performance criteria are permitted to be used when approved.

Glazing and impact-protective systems in buildings and other structures classified as Risk Category IV in accordance with Section 1.5 shall comply with the "enhanced protection" requirements of Table 3 of ASTM E1996. Glazing and impact-protective systems in all other structures shall comply with the "basic protection" requirements of Table 3 of ASTM E1996.

26.12.4 Multiple Classifications. If a building by definition complies with both the "open" and "partially enclosed" definitions, it shall be classified as an "open" building.

26.13 INTERNAL PRESSURE COEFFICIENTS

Internal pressure coefficients, (GC_{pi}), shall be determined from Table 26.13-1 based on building enclosure classifications determined from Section 26.12.

26.13.1 Reduction Factor for Large-Volume Buildings, R_i. For a partially enclosed building containing a single, unpartitioned large volume, the internal pressure coefficient, (GC_{pi}), shall be multiplied by the following reduction factor, R_i:

$$R_i = 1.0 \quad \text{or}$$

$$R_i = 0.5\left(1 + \frac{1}{\sqrt{1 + \frac{V_i}{22{,}800A_{og}}}}\right) < 1.0 \qquad (26.13\text{-}1)$$

where

A_{og} = total area of openings in the building envelope (walls and roof, in ft^2); and
V_i = unpartitioned internal volume, in ft^3.

26.14 TORNADO LIMITATION

Tornadoes have not been considered in the wind load provisions.

26.15 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

This section lists the consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

AAMA 512, *Voluntary Specifications for Tornado Hazard Mitigating Fenestration Products*, American Architectural Manufacturers Association, 2011.
Cited in: C26.14.4

ANSI A58.1, *Minimum Design Loads for Buildings and Other Structures*, American National Standards Institute, 1982.
Cited in: Section C26.5.2

ASTM E1886, *Standard test method for performance of exterior windows, curtain walls, doors, and impact protective systems impacted by missile(s) and exposed to cyclic pressure differentials*, ASTM International, 2013.
Cited in: Section 26.12.3.2, C26.12, C26.14.4.

ASTM E1996, *Standard specification for performance of exterior windows, curtain walls, doors, and impact protective systems impacted by windborne debris in hurricanes*, ASTM International, 2014.
Cited in: Section 26.12.3.2, C26.12, C26.14.4.

ANSI/DASMA 115, *Standard Method for Testing Sectional Garage Doors: Determination of Structural Performance under Missile Impact and Cyclic Wind Pressure*, Door and Access Systems Manufacturers Association International, 2005.
Cited in: Section 26.12.3.2, C26.12.

ASTM E330, *Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights, and Curtain Walls by Uniform Static Air Pressure Difference*, ASTM International, 2014.
Cited in: Section C26.5.1

CAN/CSA A123.21, *Standard test method for the dynamic wind uplift resistance of membrane-roofing systems*, CSA Group, 2014.
Cited in: Section C26.5.1

ICC 500, *ICC/NSSA Standard for the Design and Construction of Storm Shelters*, International Code Council and National Storm Shelter Association, 2014.
Cited in: Section C26.14.1, C26.14.3, C26.14.4

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